

Edition 26.09.2017

$$\frac{\partial n}{\partial t} + \frac{1}{\sqrt{s}} \frac{\partial}{\partial} \left(\frac{\sqrt{s}}{\mathbf{h}} \Gamma \right) + \frac{1}{\sqrt{s}} \frac{\partial}{\partial y} \left(\frac{\sqrt{s}}{\mathbf{h}_y} \Gamma_y \right) = n, \quad = 0, \dots, n_s \quad (1)$$

$$\Gamma = \left(\mathbf{h}_{\parallel} + {}^{(E)} + {}^{(AN)} + {}^{co-dpc} \right) n - D^n \frac{1}{\mathbf{h}} \frac{\partial n}{\partial} - \tilde{D}^p \frac{1}{\mathbf{h}} \frac{\partial p}{\partial} - D_y \frac{1}{\mathbf{h}_y} \frac{\partial n}{\partial y}$$

$$\Gamma_y = \left({}^{(E)}_y + {}^{(AN)}_y + \frac{j^{(AN)}_y + j^{(n)}_y + \tilde{j}^{(v\parallel)}_y}{en} \right) n - D^n \frac{1}{\mathbf{h}_y} \frac{\partial n}{\partial y} - \tilde{D}^p \frac{1}{\mathbf{h}_y} \frac{\partial p}{\partial y} + D_y \frac{1}{\mathbf{h}_y} \frac{\partial n}{\partial},$$

$$\Gamma = \left(\mathbf{h}_{\parallel} + {}^{(E)} + {}^{\sim(d)} + {}^{(AN)} + {}^{co-dpc} \right) n - D^n \frac{1}{\mathbf{h}} \frac{\partial n}{\partial} - \tilde{D}^p \frac{1}{\mathbf{h}} \frac{\partial p}{\partial}$$

$$\Gamma_y = \left({}^{(E)}_y + {}^{\sim(d)}_y + {}^{(AN)}_y + \frac{j^{(AN)}_y + j^{(n)}_y + \tilde{j}^{(v\parallel)}_y}{en} \right) n - D^n \frac{1}{\mathbf{h}_y} \frac{\partial n}{\partial y} - \tilde{D}^p \frac{1}{\mathbf{h}_y} \frac{\partial p}{\partial y}, \quad (2)$$

$$\Gamma^{Co} = \left(\mathbf{h}_{\parallel} + {}^{(E)} + 2 {}^{\sim(d)} + {}^{(AN)} + {}^{co-dpc} \right) n - D^n \frac{1}{\mathbf{h}} \frac{\partial n}{\partial} - \tilde{D}^p \frac{1}{\mathbf{h}} \frac{\partial p}{\partial}$$

$$\Gamma_y^{Co} = \left({}^{(E)}_y + 2 {}^{\sim(d)}_y + {}^{(AN)}_y + \frac{j^{(AN)}_y + j^{(n)}_y + \tilde{j}^{(v\parallel)}_y}{en} \right) n - D^n \frac{1}{\mathbf{h}_y} \frac{\partial n}{\partial y} - \tilde{D}^p \frac{1}{\mathbf{h}_y} \frac{\partial p}{\partial y},$$

$$\Gamma^{(no\ d\ ft)} = \left(\mathbf{h}_{\parallel} + {}^{(AN)} + 0 \cdot {}^{co-dpc} \right) n$$

$$\Gamma_y^{(no\ d\ ft)} = \left({}^{(AN)}_y \right) n, \quad e_y = 0, \quad s_y^{(AN)} = 0$$

$${}^{(E)} = \begin{cases} -\frac{B_z}{B^2 \mathbf{h}_y} \frac{\partial \phi}{\partial y}, & z \neq 0, \\ 0, & z = 0 \end{cases}, \quad {}^{(E)}_y = \begin{cases} \frac{B_z}{B^2 \mathbf{h}_y} \frac{\partial \phi}{\partial}, & z \neq 0, \\ 0, & z = 0 \end{cases},$$

$${}^{\sim(d)} = \begin{cases} \frac{B_z}{Z e \mathbf{h}_y} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right), & z \neq 0, \\ 0, & z = 0 \end{cases}, \quad {}^{\sim(d)}_y = \begin{cases} -\frac{B_z}{Z e \mathbf{h}_y} \frac{\partial}{\partial} \left(\frac{1}{B^2} \right), & z \neq 0, \\ 0, & z = 0 \end{cases},$$

$$\Gamma^{(E)} = \left(\mathbf{h}_{\parallel} + {}^{(E)} + {}^{(d)} + {}^{(AN)} + {}^{co-dpc} \right) n - D^n \frac{1}{\mathbf{h}} \frac{\partial n}{\partial}$$

$$\Gamma_y^{(E)} = \left({}^{(E)}_y + {}^{(d)}_y + {}^{(AN)}_y + \frac{j^{(AN)}_y + j^{(n)}_y + \tilde{j}^{(v\parallel)}_y}{e} \right) n - D^n \frac{1}{\mathbf{h}_y} \frac{\partial n}{\partial y}$$

where

$${}^{(d)} = \begin{cases} -\frac{B_z}{Z e B^2 n \mathbf{h}_y} \frac{\partial n}{\partial y}, & z \neq 0, \\ 0, & z = 0 \end{cases}, \quad {}^{(d)}_y = \begin{cases} \frac{B_z}{Z e B^2 n \mathbf{h}_y} \frac{\partial n}{\partial}, & z \neq 0, \\ 0, & z = 0 \end{cases},$$

$$E_{\perp} = -\frac{1}{2} \left[\mu_{\parallel}^2 + \left(\mu_{\perp}^{(d)} + \mu_{\perp}^{(E B)} \right)^2 + \left(\mu_y^{(d)} + \mu_y^{(E B)} \right)^2 \right]$$

where

$$\mu_{\perp}^{(d)} = \begin{cases} -\frac{1}{Z e B n} \frac{\partial n}{\partial y}, & z \neq 0, \\ 0, & z = 0 \end{cases}, \quad \mu_y^{(d)} = \begin{cases} \frac{B_z}{Z e B^2 n} \frac{\partial n}{\partial}, & z \neq 0, \\ 0, & z = 0 \end{cases}$$

$$\mu_{\perp}^{(E B)} = \begin{cases} -\frac{1}{B} \frac{\partial \varphi}{\partial y}, & z \neq 0, \\ 0, & z = 0 \end{cases}, \quad \mu_y^{(E B)} = \begin{cases} \frac{B_z}{B^2} \frac{\partial \varphi}{\partial}, & z \neq 0, \\ 0, & z = 0 \end{cases}$$

Source terms in density equations

$$n_s = n_{s-1} + n_{s-1} + \frac{n_{s-1}}{n_{s-1}} + \frac{n_{s-1}}{n_{s-1}} + \frac{n_{s-1}}{n_{s-1}},$$

z - the atomic charge of species ()

z_n - the nuclear charge of species ()

$$n_s = \begin{cases} -n_e n_s, & s=0, \dots, n_s-2, \quad z_{s-1} < z_{s+1} \quad \text{and} \quad z_n = z_{n+1}, \\ 0, & \end{cases}$$

$$n_{s+1} = \begin{cases} n_e n_s, & s=0, \dots, n_s-2, \quad z_{s-1} < z_{s+1} \quad \text{and} \quad z_n = z_{n+1} \\ 0, & \end{cases}$$

$$n_s = \exp \left(\mu_{0,s} + \mu_{1,s} \ln \left(\frac{e}{e} \right) \right)$$

$$n_s = \begin{cases} -n_e n_s, & s=1, \dots, n_s-1, \quad z_{s-1} < z_{s+1} \quad \text{and} \quad z_{n-1} = z_n \\ 0, & \end{cases}$$

$$n_{s-1} = \begin{cases} n_e n_s, & s=1, \dots, n_s-1, \quad z_{s-1} < z_{s+1} \quad \text{and} \quad z_{n-1} = z_n \\ 0, & \end{cases},$$

$$n_s = \exp \left(\mu_{0,s} + \mu_{1,s} \ln \left(\frac{e}{e} \right) \right)$$

$$n_{CX} = \begin{cases} -n_{0,CX} n_s, & s=1, \dots, n_s-1, \quad z_{s-1} < z_{s+1} \quad \text{and} \quad z_{n-1} = z_n \quad ??? \\ 0, & \end{cases}$$

$$n_{CX-1} = \begin{cases} n_{0,CX} n_s, & s=1, \dots, n_s-1, \quad z_{s-1} < z_{s+1} \quad \text{and} \quad z_{n-1} = z_n \quad ????? \\ 0, & \end{cases}$$

$$n_{CX} = \exp \left(\mu_{0,CX} + \mu_{1,CX} \ln \left(\frac{p}{e} \right) \right)$$

$$n_e = \sum_{n=0}^{n_e-1} z_n,$$

$$n = \sum_{n=0}^{n_e-1} n$$

$$p = n(z_e +),$$

$$p_e = n_{e_e}$$

$$c_{co-dpc} = \frac{c^{co-dpc}}{16} \left| \sqrt{\frac{p}{n}} \frac{\partial^2}{\partial^2} \left(\frac{1}{p} \frac{\partial p}{\partial} \right) \right|$$

$$\begin{cases} D_y = \left(\frac{1}{B^2} - \frac{1}{\langle B^2 \rangle} \right) \frac{B_z}{z_e}, & z \neq 0 \\ D_y = 0, & z = 0 \end{cases}, \text{ where } \langle B^2 \rangle = \frac{\iint_{co_e} B^2 \sqrt{\varepsilon} d\, dy}{\iint_{co_e} \sqrt{\varepsilon} d\, dy}$$

$$\begin{cases} D_y = \left(\frac{1}{B^2} - \frac{1}{\langle B^2 \rangle} \right) \frac{B_z}{z_e}, & z \neq 0 \\ D_y = 0, & z = 0 \end{cases}$$

$$D^n = D_{AN}^n$$

$$D_{AN}^n = \left(c_{D^N,0} + c_{D^N,1} \frac{10^{20}}{n_e} + c_{D^N,2} D^f + c_{D^N,3} \frac{6,2 \cdot 10^{-2}}{e|B|} e \right) f_D(,y),$$

$$f_D(,y) = \begin{cases} 1, & z = 0 \\ \alpha_{1,D}, & -\delta_{1,D} \leq y \leq 0, \quad \in Co\,e, \quad z \neq 0 \\ \alpha_{2,D}, & 0 \leq y \leq \delta_{2,D}, \quad \in Co\,e, \\ \alpha_{3,D}, & , y \in Pr \cup OL \end{cases}$$

$$=0, \, c_{D^N,0}=0, \, c_{D^N,1}=0, \, c_{D^N,2}=0, \, c_{D^N,3}=0$$

$$=1, \, c_{D^N,0}=0, \, c_{D^N,1}=0, \, c_{D^N,2}=1, \, c_{D^N,3}=0$$

$$D^f = c_{D^f,0} + c_{D^f,1} \frac{10^{20}}{n} + \frac{c_{D^f,2} \sqrt{E}}{c_{D^f,3} n + c_{D^f,4} n_e},$$

	$c_{D^f,0}$	$c_{D^f,1}$	$c_{D^f,2}$	$c_{D^f,3}$	$c_{D^f,4}$	$c_{D^f,5}$	$c_{D^f,6}$	$c_{D^f,7}$
0	10^2	0	1	$5 \cdot 10^{-20}$	$1,5 \cdot 10^{-18}$	0	0	0
1	0,3	0	0	—	—	0	0	0

$$\tilde{D}^p = \begin{cases} \frac{D^p}{\left(1 + \frac{\left(\frac{\sqrt{\mathcal{S}}}{\not{p}} D^p \left| \frac{1}{\not{p}} \frac{\partial p}{\partial} \right| \right)^{\gamma_1}}{\alpha \frac{\sqrt{\mathcal{S}}}{\not{p}} n \frac{1}{4}} \right)^{\frac{1}{\gamma_1}}}, & z = 0 \\ D^p, & z \neq 0 \end{cases} = \sqrt{\frac{8}{\pi}}, \quad z = 0$$

$$\tilde{D}_y^p = \begin{cases} \frac{D^p}{\left(1 + \frac{\left(\frac{\sqrt{\mathcal{S}}}{\not{p}_y} D^p \left| \frac{1}{\not{p}_y} \frac{\partial p}{\partial y} \right| \right)^{\gamma_1}}{\alpha \frac{\sqrt{\mathcal{S}}}{\not{p}_y} n \frac{1}{4}} \right)^{\frac{1}{\gamma_1}}}, & z = 0 \\ D^p, & z \neq 0 \end{cases} = \sqrt{\frac{8}{\pi}}, \quad z = 0$$

$$D^p = D_{AN}^p$$

$$D_{AN}^p = \frac{1}{z_e +} \left(c_{D^p,0} + c_{D^p,1} \frac{10^{20}}{n_e} + c_{D^p,2} D^f + c_{D^p,3} \frac{6,2 \cdot 10^{-2}}{e|B|} e \right),$$

$$= 0, \quad c_{D^p,0} = 0, \quad c_{D^p,1} = 0, \quad c_{D^p,2} = 1, \quad c_{D^p,3} = 0$$

$$= 1, \quad c_{D^p,0} = 0, \quad c_{D^p,1} = 0, \quad c_{D^p,2} = 0, \quad c_{D^p,3} = 0$$

$$^{(AN)} = 0$$

$$^{(AN)}_y = c_{^{(AN)},0} + c_{^{(AN)},1} \frac{10^{20}}{n_e} + c_{^{(AN)},2} D^f + c_{^{(AN)},3} \frac{6,2 \cdot 10^{-2}}{e|B|} e$$

$$= 0, \quad c_{^{(AN)},0} = 0, \quad c_{^{(AN)},1} = 0, \quad c_{^{(AN)},2} = 0, \quad c_{^{(AN)},3} = 0$$

$$= 1, \quad c_{^{(AN)},0} = 0, \quad c_{^{(AN)},1} = 0, \quad c_{^{(AN)},2} = 0, \quad c_{^{(AN)},3} = 0$$

New form

$$\begin{aligned} & \frac{\partial n_{\parallel}}{\partial t} + \frac{1}{h_z \sqrt{s}} \frac{\partial}{\partial} \left(\frac{h_z \sqrt{s}}{h} \Gamma \right) + \frac{1}{h_z \sqrt{s}} \frac{\partial}{\partial y} \left(\frac{h_z \sqrt{s}}{h_y} \Gamma_y \right) + \frac{b}{h} \frac{\partial n}{\partial} + Z e n \frac{b}{h} \frac{\partial \varphi}{\partial} = \\ & = \parallel + c + f + e + i + \underline{+_{cv} +_{AN}} \end{aligned}$$

Old form

$$\begin{aligned} & \frac{\partial n_{\parallel}}{\partial t} + \frac{1}{h_z \sqrt{s}} \frac{\partial}{\partial} \left(\frac{h_z \sqrt{s}}{h} \Gamma \right) + \frac{1}{h_z \sqrt{s}} \frac{\partial}{\partial y} \left(\frac{h_z \sqrt{s}}{h_y} \Gamma_y \right) + \frac{b}{h} \frac{\partial n}{\partial} + \frac{b}{h} \frac{Z n}{n_e} \frac{\partial n_{e,e}}{\partial} = \\ & = \parallel + c, + f + e + i + \underline{+_{cv} +_{AN}} \quad o d fo \end{aligned}$$

$$\begin{aligned} \parallel & = -(\nabla \pi^{\parallel})_{\parallel} \\ (\nabla \pi^{\parallel})_{\parallel} & = \begin{cases} 0, & z \neq 1 \\ \frac{\partial}{h_z \sqrt{s} \partial} \left(\frac{h_z \sqrt{s}}{h} \frac{4}{3} \eta \frac{\partial \ln \left(\frac{h_z B^{\frac{1}{2}}}{B^{\frac{1}{2}} h} \right)}{\partial} \right) B^{\frac{1}{2}}_{\parallel} - B^{\frac{3}{2}}_{\parallel} \frac{\partial}{h \partial} \left(\alpha^{NEO} \frac{b \tau}{B^2} \frac{\partial \left(B^{\frac{1}{2}} \tilde{q}^{(0)} \right)}{h \partial} \right) \end{cases}, \quad z = 1 \end{aligned}$$

$$c, = b n_{\parallel}^2 \frac{1}{h h_z} \frac{\partial}{\partial}$$

$$AN = \begin{cases} 0, & z \neq 1 \quad o \quad z_n \neq 1 \\ -j_y^{(AN)} \frac{BB}{B_z}, & z = 1 \quad nd \quad z_n = 1 \end{cases} \text{ - correcting source due to artificial anomalous electric conductivity}$$

$$= f, + e + i + \underline{+_{cv} +_{AN}}$$

Parallel momentum source due to ionization

$$i = \frac{q}{i} + \frac{q}{i+1}$$

$$i^q = \begin{cases} -i n_e n \left(\frac{+}{2} +1 \right)_{\parallel}, & = 0, \dots, n_s - 2, \quad z < z_{+1} \quad nd \quad z_n = z_{n+1}, \\ 0, & \end{cases}$$

$$i+1^q = \begin{cases} i n_e n \left(\frac{+}{2} +1 \right)_{\parallel}, & = 0, \dots, n_s - 2, \quad z < z_{+1} \quad nd \quad z_n = z_{n+1}, \\ 0, & \end{cases}$$

Parallel momentum source due to recombination

$$= \underline{\frac{q}{i} + \frac{q}{i+1}}$$

$$q = \begin{cases} - \frac{n n \left(\frac{+}{-} - 1 \right)}{2} \parallel, & = 1, \dots, n_s - 1, \quad z_{-1} < z \quad nd \quad z_{n-1} = z_n \\ 0 \end{cases}$$

$$q = \begin{cases} \frac{n n \left(\frac{+}{-} - 1 \right)}{2} \parallel, & = 1, \dots, n_s - 1, \quad z_{-1} < z \quad nd \quad z_{n-1} = z_n \\ 0, \end{cases}$$

$$CX =$$

New friction

$$f = f_{f,e} + f_s,$$

$$f_{f,e} = \frac{b^2 e^2 z^2}{\sigma_{\parallel} z_{eff}} n_e n \left(\epsilon_{\parallel} - \epsilon_{\parallel} \right) \quad z \neq 0$$

$$f_s = \sum_{\substack{b=0 \\ b \neq}}^{n_s-1} \frac{1}{\zeta_p} \frac{1}{\sqrt{\frac{p}{p} + \frac{b}{p}}} \frac{p}{p} z^2 z_b^2 n_b n \left(\epsilon_{\parallel} - \epsilon_{\parallel} \right)$$

$$\zeta_p = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{p}}{\Lambda} \left(\frac{4\pi\epsilon_0}{e^2} \right)^{\frac{3}{2}} \quad \tau = \frac{\zeta_p}{z^2 z_{eff} n_e} \sqrt{\frac{p}{p}}$$

$$\epsilon_s^{(1)} = \begin{cases} c_p^{(1)}, & = n, \quad s \neq, \quad z \neq 0, \quad z_s \neq 0, \\ c_p^{(1)}, & s = s_n, \quad \neq s, \quad z \neq 0, \quad z_s \neq 0, \\ 1, & \neq n, \quad s \neq s_n, \quad z \neq 0, \quad z_s \neq 0, \\ 0, & = s, \quad o \quad z = 0, \quad z_s = 0, \end{cases}$$

$$c_p^{(1)} = \frac{(1+0.24z_{eff-p})(1+0.93z_{eff-p})}{(1+2.56z_{eff-p})(1+0.29z_{eff-p})} \quad c_p^{(2)} = 1.56 \frac{(1+1.4z_{eff-p})(1+0.52z_{eff-p})}{(1+2.56z_{eff-p})(1+0.29z_{eff-p})}$$

$$z_{eff-p} = \frac{\sum_{\substack{b=0 \\ b \neq}}^{n_s-1} n z^2}{n z_n^2}$$

Old friction

$$f_s = \sum_{\substack{b=0 \\ b \neq, z \neq 0, z_b \neq 0}}^{n_s-1} \frac{1}{\tau_p} \frac{p}{p} \sqrt{\frac{p}{p} + \frac{b}{p}} z^2 z_b^2 n_b n \left(\epsilon_{\parallel} - \epsilon_{\parallel} \right), \text{ - friction force (b2sifr)}$$

$$\tau_p = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{p}}{\Lambda} \left(\frac{4\pi\epsilon_0}{e^2} \right)^{\frac{3}{2}}, \text{ where the Coulomb logarithm } \Lambda \text{ can be chosen as}$$

$$\Lambda = \begin{cases} 12 \\ \max \left\{ 5; 23,4 - 1,15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) 3,45 \cdot \log_{10} \left(\frac{e}{e} \right) \right\}, & \frac{e}{e} \leq 50 \\ \max \left\{ 5; 25,3 - 1,15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) 2,30 \cdot \log_{10} \left(\frac{e}{e} \right) \right\}, & \frac{e}{e} > 50 \\ \max \left\{ 5; 17,3 - 0,5 \cdot \log_e \left(\frac{n_e}{10^{20}} \right) 1,5 \cdot \log_e \left(\frac{e}{e \cdot 10^3} \right) \right\} & \text{Wesson formula for ion-ion collisions} \end{cases} \quad \text{Braginsky formula},$$

New thermal force

$$e = e_{,e} + e_{,e},$$

$$e_{,e} = -\alpha_e c_e^{(1)} \frac{n z^2}{b e \zeta_e} \frac{1}{c_{\sigma_{CL}}^{(b c n)}} \frac{1}{b} \frac{\partial e}{\partial}, \quad z \neq 0$$

$$e_{,e} = \sum_{\substack{p=0 \\ p \neq}}^{n_s-1} c_{\alpha_p}^{(\lim)} z^2 z_p^2 \frac{p}{\zeta_p e n + n_p} \alpha_p \frac{1}{b} \frac{\partial}{\partial}, \quad z \neq 0$$

$$\alpha_p = \frac{\zeta_p e(n + n_p)}{\sqrt{\frac{p}{p}}} \sqrt{\frac{\frac{p}{p}}{\frac{p}{p} + \frac{p}{p}}} \left(\frac{\frac{e}{p}}{\kappa_p} - \frac{\frac{e}{p}}{\kappa} \right) b, \quad z \neq 0$$

$$\zeta_e = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{e}}{\Lambda} \frac{e^{\frac{3}{2}}}{e^2} \left(\frac{4\pi\epsilon_0}{e^2} \right)^2 \quad \tau_e = \frac{\zeta_e}{z_{eff} n_e}$$

$$c_e^{(1)} = \frac{(1+0.24z_{eff})(1+0.93z_{eff})}{(1+2.56z_{eff})(1+0.29z_{eff})}, \quad c_e^{(2)} = 1.56 \frac{(1+1.4z_{eff})(1+0.52z_{eff})}{(1+2.56z_{eff})(1+0.29z_{eff})} \quad \tilde{c}_e^{(2)} = \frac{c_e^{(2)}}{\left(z_{eff} + \frac{\sqrt{2}}{2} \right)}$$

$$\left(\frac{e}{s} \right) = \begin{cases} c_p^{(2)}, & s \neq, \quad s = s_n, \quad z \neq 0, z_s \neq 0, \\ 0, & \neq n, \quad s \neq s_n, \quad z \neq 0, z_s \neq 0, \\ 0, & = n, \quad s - \text{arbitrary} \end{cases}$$

$$\kappa = \sum_{p=0}^{n_s-1} z^2 z_p^2 n \sqrt{\frac{p}{p}} \sqrt{\frac{\frac{p}{p}}{\frac{p}{p} + \frac{p}{p}}}$$

$$c_{\alpha_p}^{(\lim)} = \begin{cases} 1, & \text{inside the separatrix} \\ \frac{1}{\alpha_p \frac{1}{b} \frac{\partial}{\partial}}, & \text{outside the separatrix} \\ 1 + \frac{\alpha_p \frac{1}{b} \frac{\partial}{\partial}}{c_{f \lim} j_{p \max}}, & \end{cases}$$

$$j_{p \max} = e \left(\frac{e}{p} \right) \sqrt{\frac{p}{p}} (n + n_p)$$

New coefficient $c_{f \lim p}$ is stored in cflim(4)

Old thermal force

$$e_{\parallel} = n \left(\frac{z}{n_e} - \frac{z^2}{n_{e^2}} \right) n_e \mathbf{b} \cdot e \left[\hat{E} - \frac{1}{e} (c_{\mathbf{b} \cdot e}, \Delta_e + c_{\mathbf{b} \cdot \Delta} \right], \text{ thermal force (b2sifr)}$$

$$n_{e^2} = \sum_{n=0}^{n_s-1} z^2 n, \quad t_e = 0,3; \quad t = 0,3; \quad z_{\lambda_e} = \frac{1.5 \cdot 10^{16}}{n_e} \left(\frac{-e}{e} \right)^2, \quad z_{\lambda} = \frac{1.5 \cdot 10^{16}}{n_{e^2}} \left(\frac{-e}{e} \right)^2$$

$$\Delta_e = \begin{cases} \frac{\partial}{\partial t_e}, & t_e \leq 0 \\ \left| \min \left\{ \frac{\partial}{\partial t_e}, \frac{t_e}{z_{\lambda_e} |\mathbf{b}|} \right\} \right|, \frac{\partial}{\partial t_e} \geq 0, & t_e > 0 \\ - \left| \min \left\{ \frac{\partial}{\partial t_e}, \frac{t_e}{z_{\lambda_e} |\mathbf{b}|} \right\} \right|, \frac{\partial}{\partial t_e} < 0, & t_e > 0 \end{cases}$$

$$\Delta = \begin{cases} \frac{\partial}{\partial t}, & t \leq 0 \\ \left| \min \left\{ \frac{\partial}{\partial t}, \frac{t}{z_{\lambda} |\mathbf{b}|} \right\} \right|, \frac{\partial}{\partial t} \geq 0, & t > 0 \\ - \left| \min \left\{ \frac{\partial}{\partial t}, \frac{t}{z_{\lambda} |\mathbf{b}|} \right\} \right|, \frac{\partial}{\partial t} < 0, & t > 0 \end{cases}$$

$c_{\mathbf{b} \cdot e} = 0, \quad = 0, n_s - 1$ and $c_{\mathbf{b} \cdot \Delta} = 2,65, \quad = 0, n_s - 1$ if 21 moment approximation is not used.

$$\Gamma = \begin{cases} \parallel \Gamma^{Co} + \frac{4}{3} \eta \frac{\partial \ln \mathbf{b} \cdot z}{\partial} \parallel - \frac{4}{3} \eta \frac{\partial \parallel}{\partial}, & z = 1 \\ \parallel \Gamma^{Co} - \frac{4}{3} \eta \frac{\partial \parallel}{\partial}, & z \neq 1 \end{cases}$$

$$\Gamma_y = \parallel \Gamma_y^{Co} - \eta_y \frac{\partial \parallel}{\partial y},$$

$$\eta = \eta_{AN} + \tilde{\tilde{\eta}}_{CL}$$

$$\eta_{AN} = \begin{cases} \left(c_{\eta,0} + c_{\eta,1} \frac{10^{20}}{n_e} + c_{\eta,2} D^f + c_{\eta,3} \frac{6,2 \cdot 10^{-2}}{e|B|} e \right) n f_{\eta}(\cdot, y), & z \neq 0 \\ \left(c_{\eta,0} + c_{\eta,1} \frac{10^{20}}{n_e} + c_{\eta,2} D^f + c_{\eta,3} \frac{6,2 \cdot 10^{-2}}{e|B|} e \right) n, & z = 0 \end{cases}$$

$$f_{\eta}(\cdot,y)=\begin{cases}1, & z=0 \\ \alpha_{1,\eta}, & -\delta_{1,\eta}\leq y\leq 0, \quad \in Coe, \quad z\neq 0 \\ \alpha_{2,\eta}, & 0\leq y\leq \delta_{2,\eta}, \quad \in Coe, \\ \alpha_{3,\eta}, & ,y\in Pr\cup OL\end{cases}$$

$$=0, c_{\eta,0}=0, c_{\eta,1}=0, c_{\eta,2}=1, c_{\eta,3}=0$$

$$=1, c_{\eta,0}=0, c_{\eta,1}=0, c_{\eta,2}=1, c_{\eta,3}=0$$

$$\tilde{\eta}_{CL}=c_{\eta^{(CL)}}\tilde{\eta}_{CL}$$

$$c_{\eta^{(CL)}}=\begin{cases}1, & z\neq 0 \text{ nd on y fo nne } f_u \text{ \& } f \text{ ces} \\ \frac{1}{\frac{\sqrt{\delta^*}}{h}\eta_{CL}\left|\frac{\partial}{\partial}\right|_{\parallel}}}, & \text{fo } o h e s c ses\end{cases}$$

$$^{(max)}=c_{f\lim_2}\frac{\sqrt{\delta^*}}{h}\left|h\right|n$$

$$\eta_{CL}=c_{\eta^{(CL)}}^{(l_u^{c\ n})}\eta_{CL}$$

‘b2trcl_lluciani’ ‘1’

$$c_{\eta^{(CL)}}^{(l_u^{c\ n})}=\begin{cases}\frac{1}{7,5\cdot 10^{16}\left(\frac{-}{e}\right)^2}, & z\neq 0 \text{ nd on y fo nne } f_u \text{ \& } f \text{ ces}, \\ 1+\frac{c(y)\sum_{n=0}^{n_s-1}z^2n}{}, & \text{fo } o h e s c ses\end{cases}$$

For plateau and banana correction ‘b2trcl_lluciani’ ‘3’

$$c_{\eta^{(CL)}}^{(l_u^{c\ n})}=\begin{cases}\frac{1}{15,12\cdot 10^{16}\left(\frac{-}{e}\right)^2}\cdot\frac{1}{15,12\cdot 10^{16}\cdot \epsilon^{3/2}\left(\frac{-}{e}\right)^2}, & z\neq 0 \text{ nd on y fo nne } f_u \text{ \& } f \text{ ces}, \\ 1+\frac{c(y)\sum_{n=0}^{n_s-1}z^2n}{c(y)\sum_{n=0}^{n_s-1}z^2n}, & \text{fo } o h e s c ses\end{cases}$$

$$c(y)=\frac{h}{h}\left|h\right|^{-1}d\ , \qquad \epsilon=\frac{\left(\frac{h}{h}d\right)^2}{\oint\sqrt{\delta^*}h_yd}$$

$$\eta_{CL} = \begin{cases} 0, & z = 0 \\ \eta \sqrt{2} n \tau b^2, & z \neq 0 \end{cases}, \eta = 0.96,$$

Old expression was $\eta_{CL} = \begin{cases} 0, & z = 0 \\ \eta n \tau b^2, & z \neq 0 \end{cases}, \eta = 1.357,$

$$\tau = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{\frac{3}{2}}}{z^2 z_{eff} n_e \Lambda} \left(\frac{4\pi\epsilon_0}{e^2} \right)^2, \text{ where the Coulomb logarithm } \Lambda \text{ can be chosen as}$$

$$\Lambda = \begin{cases} 12 \\ \max \left\{ 5; 23.4 - 1.15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) \right. \left. 3.45 \cdot \log_{10} \left(\frac{e}{e} \right) \right\}, & \frac{e}{e} \leq 50 \\ \max \left\{ 5; 25.3 - 1.15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) \right. \left. 2.30 \cdot \log_{10} \left(\frac{e}{e} \right) \right\}, & \frac{e}{e} > 50 \end{cases} \quad \begin{array}{l} \text{Braginsky formula} \\ \text{Wesson formula for ion-ion collisions} \end{array},$$

$$\eta_y = \eta_{AN} + \eta_{CL,y}, \quad \eta_y^{(CL)} = 0$$

$$\tilde{q}^{(0)} = q^{(1)} - c_{q1} \left(\frac{q^{(0)}}{q^{(1)}} \right), \quad q^{(1)} = c_{q1} n \sqrt{\frac{+}{e}}$$

$$c_q = 0.5 \quad \text{or} \quad \{ 'b2siav_cqip1' \quad '0.1' \} \text{ in b2mn.dat}$$

the three-point smoothing procedure is imposed twice on $\tilde{q}^{(0)}$ along $-$ direction

in case if $\{ 'b2siav_style_qip' \quad '0' \}$ in b2mn.dat

$$q^{(0)} = q_{||}^{(0)} + \frac{B}{B} q^{(d)} = \begin{bmatrix} 0 \\ -\frac{5}{2} \frac{B_z B}{B^2} \frac{n}{e} \frac{\partial}{\partial y} \end{bmatrix} \quad \begin{array}{l} q_{||} \text{ ts de } e \text{ sep } t \\ ns \text{ de } e \text{ sep } t \end{array}$$

in case if $\{ 'b2siav_style_qip' \quad '1' \}$ in b2mn.dat

$$q^{(0)} = q_{||}^{(0)} + \frac{B}{B} q^{(d)} = \begin{bmatrix} 0 \\ -\tilde{\approx}_{(CL)} \frac{\partial}{\partial y} - \frac{5}{2} \frac{B_z B}{B^2} \frac{n}{e} \frac{\partial}{\partial y} \end{bmatrix} \quad \begin{array}{l} q_{||} \text{ ts de } e \text{ sep } t \\ ns \text{ de } e \text{ sep } t \end{array}$$

$$\tilde{\tau} = \frac{\tau}{1 + \frac{15,12 \cdot 10^{16} \left(\frac{+}{e} \right)^2}{c(y) \sum_{n=0}^{n_s-1} z^2 n}} \cdot \frac{1}{1 + \frac{15,12 \cdot 10^{16} \cdot \epsilon^{3/2} \left(\frac{+}{e} \right)^2}{c(y) \sum_{n=0}^{n_s-1} z^2 n}}$$

This correction is compatible **only** with plateau and banana correction $'b2trcl_lluciani' \quad '3'$

$\alpha^{NEO} = \frac{8}{15} \left(\frac{-0.17 + 1.05\nu_*^{1/2} + 2.7\nu_*^2 \varepsilon^3}{1 + 0.7\nu_*^{1/2} + \nu_*^2 \varepsilon^3} - 1 \right)$ gives the correct coefficient before the temperature gradient in the neoclassical electric field in the plateau and banana regime

$$\nu_* = \frac{c(y)}{2\pi \varepsilon^{3/2} \tau \sqrt{I}}, \quad c(y) = \oint \frac{1}{|b|} dy, \quad \varepsilon = \frac{\left(\oint \frac{1}{\sqrt{s}} dy \right)^2}{\oint \sqrt{s} dy}$$

Old form

$$\begin{aligned} & \frac{3}{2} \frac{\partial n_{e,e}}{\partial t} + \frac{1}{\sqrt{s}} \frac{\partial}{\partial} \left(\frac{\sqrt{s}}{h} \tilde{q}_e \right) + \frac{1}{\sqrt{s}} \frac{\partial}{\partial y} \left(\frac{\sqrt{s}}{h_y} \tilde{q}_{ey} \right) + \frac{n_{e,e}}{\sqrt{s}} \frac{\partial}{\partial} \left(\frac{\sqrt{s}}{h} b_{ell} \right) = \\ & = n_e + c_{E \times B} n_e B_z \frac{1}{h_y} \left(\frac{\partial \phi}{\partial y} \frac{\partial}{\partial} \left(\frac{1}{B^2} \right) - \frac{\partial \phi}{\partial} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right) \right) - \frac{j_y^{(e)}}{en_e h_y} \frac{\partial n_{e,e}}{\partial y} + j \cdot \hat{E} + 0 \cdot j_y \hat{E}_y, \end{aligned} \quad (3)$$

New form is used when b2sig_style is 2

$$\begin{aligned} & \frac{3}{2} \frac{\partial n_{e,e}}{\partial t} + \frac{1}{\sqrt{s}} \frac{\partial}{\partial} \left(\frac{\sqrt{s}}{h} \tilde{q}_e \right) + \frac{1}{\sqrt{s}} \frac{\partial}{\partial y} \left(\frac{\sqrt{s}}{h_y} \tilde{q}_{ey} \right) + \frac{n_{e,e}}{\sqrt{s}} \frac{\partial}{\partial} \left(\frac{\sqrt{s}}{h} b_{ell} \right) = \\ & = n_e + c_{E \times B} n_e B_z \frac{1}{h_y} \left(\frac{\partial \phi}{\partial y} \frac{\partial}{\partial} \left(\frac{1}{B^2} \right) - \frac{\partial \phi}{\partial} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right) \right) - \frac{j_y^{(e)}}{en_e h_y} \frac{\partial n_{e,e}}{\partial y} + \dots, \end{aligned} \quad (3)$$

$$n_e = n_{ell} \sum_{\substack{z=0 \\ z \neq 0}}^{n_s-1} \left(n_{f,e} + n_{e,e} \right)$$

$$n_e = - \Delta - \dots$$

$$\Delta = c_{eqp} \Lambda e^{\frac{3}{2} \sum_{p=0}^{n_s-1} \frac{z^2}{n}} \frac{n_e n}{n^{\frac{3}{2}}} \left(n_e - \dots \right), \quad c_{eqp} = 4,8 \cdot 10^{-15}, \quad \text{where } \Lambda \text{ can be chosen as}$$

$$\Lambda = \begin{cases} 12 \\ \max \left\{ 5; 23,4 - 1,15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) \quad 3,45 \cdot \log_{10} \left(\frac{n_e}{e} \right) \right\}, & \frac{n_e}{e} \leq 50 \\ \max \left\{ 5; 25,3 - 1,15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) \quad 2,30 \cdot \log_{10} \left(\frac{n_e}{e} \right) \right\}, & \frac{n_e}{e} > 50 \end{cases} \quad \text{Braginsky formula},$$

$$\left\{ \max \left\{ 5; 15,2 - 0,5 \cdot \log_e \left(\frac{n_e}{10^{20}} \right) \quad \log_e \left(\frac{n_e}{e \cdot 10^3} \right) \right\} \right\} \quad \text{Wesson formula for electron - ion collisions}$$

$$\dots = \sum_{p=0}^{n_s-1} n_e n \quad \text{from b2stel}$$

$$\tilde{q}_e = \tilde{f}_e - \kappa_e \frac{1}{\hbar} \frac{\partial}{\partial} \frac{e}{e} + 0 \cdot \alpha_e \hat{E} - q_e^{P. \hbar}$$

New

$$\tilde{f}_e = \frac{3}{2} \Gamma_e + {}_e^{(t)} n_e - \tilde{\epsilon}_e^{(2)} \frac{j^{(\parallel)}}{e}$$

Old

$$\tilde{f}_e = \frac{3}{2} \Gamma_e + {}_e^{(t)} n_e - 0.71 \frac{j^{(\parallel)}}{e}$$

$$q_e^{P. \hbar} = -\frac{5}{2} D_{ey} \frac{1}{\hbar} \frac{\partial n_e}{\partial y}$$

$$D_{ey} = \left(\frac{1}{B^2} - \frac{1}{\langle B^2 \rangle} \right) \frac{B_z}{e} \quad D_{ey} = \left(\frac{1}{B^2} - \frac{1}{\langle B^2 \rangle} \right) \frac{B_z}{e} \quad \text{where } \langle B^2 \rangle = \frac{\iint B^2 \sqrt{s} d dy}{\iint \sqrt{s} d dy}$$

New

$$q_e = \left[\frac{3}{2} \Gamma_e - \tilde{\epsilon}_e^{(2)} \frac{j^{(\parallel)}}{e} - \frac{5}{2} n_e \frac{B_z}{e \hbar} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right) + {}_e^{(t)} n_e \right] - \kappa_e \frac{1}{\hbar} \frac{\partial}{\partial} \frac{e}{e} + 0 \cdot \alpha_e \hat{E}$$

Old

$$q_e = \left[\frac{3}{2} \Gamma_e - 0.71 \frac{j^{(\parallel)}}{e} - \frac{5}{2} n_e \frac{B_z}{e \hbar} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right) + {}_e^{(t)} n_e \right] - \kappa_e \frac{1}{\hbar} \frac{\partial}{\partial} \frac{e}{e} + 0 \cdot \alpha_e \hat{E}$$

$$\Gamma_e = \sum_{=0}^{n_s-1} z \Gamma^{\hbar e)} - \frac{j^{(\parallel)}}{e},$$

$$\Gamma^{\hbar e)} = (\hbar_{\parallel} + {}^{(E)} + \frac{5}{3} {}^{(AN)} + {}^{co-dpc}) n - \frac{5}{3} \left[D^n \frac{1}{\hbar} \frac{\partial n}{\partial} + \tilde{D}^p \frac{1}{\hbar} \frac{\partial p}{\partial} \right]$$

New

$$q_e^{(E)} = \left[\frac{5}{2} \Gamma_e^{(E)} + {}_e^{(t)} n_e - \tilde{\epsilon}_e^{(2)} \frac{\tilde{j}_{\parallel}}{e} \right] - \kappa_e \frac{1}{\hbar} \frac{\partial}{\partial} \frac{e}{e}$$

Old

$$q_e^{(E)} = \left[\frac{5}{2} \Gamma_e^{(E)} + {}_e^{(t)} n_e - 0.71 \frac{\tilde{j}_{\parallel}}{e} \right] - \kappa_e \frac{1}{\hbar} \frac{\partial}{\partial} \frac{e}{e}$$

$$q_{ey}^{(E)} = \left[\frac{5}{2} \Gamma_{ey}^{(E)} + {}_{ey}^{(t)} n_e \right] - \kappa_{ey} \frac{1}{\hbar} \frac{\partial}{\partial y} \frac{e}{e}$$

where

$$\Gamma_e^{(E)} = \left(\mathcal{B}_{\parallel e} + \mathcal{E}_e^{(E)} + \mathcal{E}_e^{(d)} + \mathcal{E}_e^{(AN)} \right) n_e - D^n \mathcal{H}_y \frac{1}{\partial} \frac{\partial n_e}{\partial}$$

$$\Gamma_{ey}^{(E)} = \left(\mathcal{E}_{ey}^{(E)} + \mathcal{E}_{ey}^{(d)} + \mathcal{E}_{ey}^{(AN)} \right) n_e - D^n \mathcal{H}_y \frac{1}{\partial y} \frac{\partial n_e}{\partial y}$$

$$\mathcal{E}_e^{(d)} = \frac{B_z}{eB^2 n_e \mathcal{H}_y} \frac{\partial n_{e,e}}{\partial y} \qquad \mathcal{E}_{ey}^{(d)} = -\frac{B_z}{eB^2 n_e \mathcal{H}_y} \frac{\partial n_{e,e}}{\partial}$$

$$\mathcal{E}_e^{(t)} = 0$$

$$\kappa_e = \kappa_e^{(AN)} + \tilde{\kappa}_e^{(CL)},$$

$$\kappa_e^{(AN)} = \chi_e n_e,$$

$$\chi_e = \left(\alpha_{e,0} + \alpha_{e,1} \frac{10^{20}}{n_e} + \alpha_{e,3} \frac{6,2 \cdot 10^{-2} e}{e|B|} \right) f_{\mathcal{H}_e}(\cdot, y),$$

$$\alpha_{e,0} = 1.2, \alpha_{e,1} = 0, \alpha_{e,3} = 0,$$

$$f_{\mathcal{H}_e}(\cdot, y) = \begin{cases} \alpha_{1\mathcal{H}_e}, & -\delta_{1\mathcal{H}_e} \leq y \leq 0, \quad \in Co\ e, \\ \alpha_{2\mathcal{H}_e}, & 0 \leq y \leq \delta_{2\mathcal{H}_e}, \quad \in Co\ e, \\ \alpha_{3\mathcal{H}_e}, & \quad, y \in Pr \cup OL \end{cases}$$

$$\tilde{\kappa}_e^{(CL)} = c_e^{(\lim)} \tilde{\kappa}_e^{(CL)}$$

$$c_e^{(\lim)} = \begin{cases} 1, & \text{on y fo nne fu \& f ces} \\ \frac{1}{\frac{\sqrt{\mathcal{S}}}{\mathcal{H}_y} \tilde{\kappa}_e^{(CL)} \left| \frac{\partial_e}{\partial} \right|}, & \text{fo o\&e fu \& f ces} \\ 1 + \frac{(\max)}{\mathcal{H}_e^{(\max)} n_{e,e}} \end{cases}$$

$$\mathcal{H}_e^{(\max)} = c_{f \lim_0} \frac{\sqrt{\mathcal{S}}}{\mathcal{H}_y} |\mathcal{B}| \sqrt{\frac{e}{e}},$$

$$\tilde{\kappa}_e^{(CL)} = c_{\kappa_e^{(CL)}}^{(I_{\mathcal{H}}^{c\ n})} \kappa_e^{(CL)} \mathcal{B}^2,$$

For plateau and banana correction ‘b2trcl_lluciani’ ‘3’

$$c_{\kappa_e^{(CL)}}^{(I_{\mathcal{H}}^{c\ n})} = \begin{cases} 0.58 \varepsilon^{3/2} / \mathcal{I}_1, & \text{only for inner flux surfaces} \\ 1, & \text{for other flux surfaces} \end{cases}$$

$$\mathcal{I}_1 = \frac{1}{1 + 2.01 \cdot \nu_{*e}^{1/2} + 1.53 \cdot \nu_{*e}} + \frac{0.52 \cdot \varepsilon^3 \cdot \nu_{*e}}{1 + 0.89 \cdot \varepsilon^{3/2} \cdot \nu_{*e}}$$

$$\nu_{*e} = \frac{c(y)}{2\pi\epsilon^{3/2}\tau_e\sqrt{\frac{l}{e}}},$$

$$\tau_e = \frac{3}{4\sqrt{2\pi}} \sqrt{\frac{e}{e^2}}^{\frac{3}{2}} \left(\frac{4\pi\epsilon_0}{e^2} \right)^2, \text{ where Coulomb logarithm } \Lambda \text{ can be chosen as}$$

$$\Lambda = \begin{cases} 12 \\ \max \left\{ 5; 23,4 - 1,15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) \right. \left. 3,45 \cdot \log_{10} \left(\frac{e}{e} \right) \right\}, & \frac{e}{e} \leq 50 \\ \max \left\{ 5; 25,3 - 1,15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) \right. \left. 2,30 \cdot \log_{10} \left(\frac{e}{e} \right) \right\}, & \frac{e}{e} > 50 \\ \max \left\{ 5; 14,9 - 0,5 \cdot \log_e \left(\frac{n_e}{10^{20}} \right) \right. \left. \log_e \left(\frac{e}{e \cdot 10^3} \right) \right\} \end{cases} \quad \begin{array}{l} \text{Braginsky formula} \\ \\ \text{Wesson formula for electron - electron collisions} \end{array},$$

but it was as below

$$c_{\kappa_e^{(CL)}}^{(h^c n)} = \begin{cases} \frac{1}{7,5 \cdot 10^{16} \left(\frac{e}{e} \right)^2}, & \text{only for inner flux surfaces} \\ 1 + \frac{c(y) \sum_{n=0}^{n_s-1} z^2 n}{c(y) \sum_{n=0}^{n_s-1} z^2 n} \\ 1, & \text{for other flux surfaces} \end{cases}$$

$$c(y) = \begin{cases} \int \frac{h^c(y)}{|h|} d, & \text{for inner flux surfaces} \\ 0, & \text{for other flux surfaces} \end{cases}$$

$$\kappa_e^{(CL)} = \frac{5}{2} \frac{e}{e} f_e(z_{eff}) n_e \tau_e, \text{ old form}$$

$$f_e(z_{eff}) = \frac{2,16 z_{eff}}{1 + 0,27 z_{eff}}, \text{ old form}$$

$$\kappa_e^{(CL)} = \frac{e}{e} f_e(z_{eff}) n_e \tau_e, \text{ new form}$$

$$f_e(z_{eff}) = \frac{3,9 + 2,3/z_{eff}}{0,31 + 1,2/z_{eff} + 0,41/z_{eff}^2}, \text{ new Zhdanov's expression}$$

$$z_{eff} = \frac{\sum_{n=0}^{n_s-1} z^2 n}{n_e},$$

$$j_{e\parallel} = \frac{1}{en_e} \left[\sum_{n=0}^{n_e-1} z_n en_{\parallel} - \frac{j^{(II)}}{b} \right],$$

New expression of parallel current

$$j^{(II)} = \sigma_{\parallel} \left(\frac{1}{en_e b} \frac{\partial n_{e\parallel}}{\partial} - \frac{1}{b} \frac{\partial \varphi}{\partial} \right) - \alpha_e \frac{1}{b} \frac{\partial}{\partial} - \frac{eb}{z_{eff}} \sum_{n=0}^{n_e-1} n_{\parallel} (z_n^2 - z_n z_{eff})$$

Old

$$j^{(II)} = \sigma_{\parallel} \hat{E} - \alpha_e \frac{1}{b} \frac{\partial}{\partial},$$

$$\hat{E} = \frac{1}{en_e b} \frac{\partial n_{e\parallel}}{\partial} - \frac{1}{b} \frac{\partial \varphi}{\partial},$$

$$\sigma_{\parallel} = \sigma_{AN\parallel} + \sigma_{CL\parallel},$$

$$\sigma_{AN\parallel} = 0$$

$$\tilde{\sigma}_{CL\parallel} = c_{\sigma_{CL\parallel}}^{(I_{uc} n)} \sigma_{CL\parallel}$$

For plateau and banana correction 'b2trcl_lluciani' '3'

$$c_{\sigma_{CL\parallel}}^{(I_{uc} n)} = \begin{cases} 0.58 \varepsilon^{3/2} / \varepsilon_1, & \text{only for inner flux surfaces} \\ 1, & \text{for other flux surfaces} \end{cases}$$

Old form

$$c_{\sigma_{CL\parallel}}^{(I_{uc} n)} = \frac{1}{1 + \frac{7.5 \cdot 10^{16} \left(\frac{e}{e} \right)^2}{c(y) \sum_{n=0}^{n_e-1} z_n^2 n}}$$

New form

$$\sigma_{CL\parallel} = b^2 \frac{e^2}{e} f_{\sigma_e}(z_{eff}) n_e \tau_e z_{eff}, \quad f_{\sigma_e}(z_{eff}) = \frac{1}{c_e^{(1)} z_{eff}}$$

Old form

$$\sigma_{CL\parallel} = b^2 \frac{e^2}{e} f_{\sigma_e}(z_{eff}) n_e \tau_e, \quad f_{\sigma_e}(z_{eff}) = \frac{4.95 z_{eff}}{1 + 1.54 z_{eff}},$$

$$\alpha_e = \alpha_{AN} + \tilde{\alpha}_{CL},$$

$$\alpha_{AN} = 0,$$

$$\tilde{\alpha}_{CL} = c_e^{(\lim)} \tilde{\alpha}_{CL}$$

$\alpha_{CL} = c^{(l_u^c \ n)}_{(CL)_e} \alpha_{CL}$

New form

$\tilde{\alpha}_{CL} = c^{(\lim)}_{\alpha_e} \tilde{\alpha}_{CL} \ , \ \ \tilde{\alpha}_{CL} = c^{(l_u^c \ n)}_{\alpha_{CL}} \alpha_{CL} \ , \ \ \alpha_{CL} = b^2 \frac{e}{e} f_{\alpha}(z_{eff}) \tau_e n_e z_{eff} \ , \ \ f_{\alpha}(z_{eff}) = - \frac{\tilde{\alpha}_e^{(2)}}{c_e^{(1)}}$

$c_{\alpha_e}^{(\lim)} = \begin{cases} 1, & \text{inside the separatrix} \\ \frac{1}{1 + \frac{\alpha_{CL} \frac{1}{\hbar} \frac{\partial}{\partial}}}{c_{\text{flime}} j_{e \max}}, & \text{outside the separatrix} \end{cases}$

$j_{e \max} = |b| en_e \sqrt{\frac{e}{e}}$

New coefficient c_{flime} is stored in cflim(3)

Old form

$\alpha_{CL} = b^2 \sqrt{\frac{5}{2}} \frac{e}{e} f_{\alpha}(z_{eff}) n_e \tau_e \ , \ \ \ f_{\alpha}(z_{eff}) = - \frac{1,25 z_{eff}}{1 + 0,41 z_{eff}} \ ,$

$\tilde{q}_{ey} = \tilde{f}_{ey \ e} - \kappa_{ey} \frac{1}{\hbar} \frac{\partial}{\partial y} \frac{e}{e} + 0 \cdot \alpha_{ey \ e} \hat{E}_y - q_{ey}^{P. \ \hbar}$

$\tilde{f}_{ey} = \frac{3}{2} \left(\Gamma_{ey} - \frac{5}{3} \frac{j_y^{(\)}}{e} \right) + \frac{(\ t \)}{ey} n_e$

$q_{ey}^{P. \ \hbar} = \frac{5}{2} D_{ey} \frac{1}{\hbar} \frac{\partial n_e^2}{\partial}$

(

— —

$$\kappa_e^{(\cdot)} = \begin{cases} 1.6\alpha^{(\cdot)}\sigma^{(\cdot)}_e/e^2, & y \in [-\Delta^{(\cdot)}, 0] \\ 0, & y \notin [-\Delta^{(\cdot)}, 0] \end{cases}, \quad \alpha^{(\cdot)} \approx 2$$

$$\alpha_{ey} = \alpha_{ANy} + \alpha_{CLy},$$

$$\alpha_{ANy} = \left(c_{\alpha,0} + c_{\alpha,1} \frac{10^{20}}{n_e} + c_{\alpha,3} \frac{6,2 \cdot 10^{-2}}{e|B|} e \right) n_e \sqrt{\frac{e}{e}}$$

$$c_{\alpha,0} = 10^{-3}, \; c_{\alpha,1} = 0, \; c_{\alpha,3} = 0$$

$$\alpha_{CLy} = 0$$

$$\begin{aligned} & \frac{3}{2} \frac{\partial n}{\partial t} + \frac{1}{\sqrt{s}} \frac{\partial}{\partial} \left(\frac{\sqrt{s}}{h} \tilde{q} \right) + \frac{1}{\sqrt{s}} \frac{\partial}{\partial y} \left(\frac{\sqrt{s}}{h_y} \tilde{q}_y \right) + \sum_{=0}^{n_s-1} \frac{n}{\sqrt{s}} \frac{\partial}{\partial} \left(\frac{\sqrt{s}}{h} \parallel \right) = \\ & = \Delta + \dots + c_{E \times B} B \frac{1}{h_y} \left(\frac{\partial \phi}{\partial y} \frac{\partial}{\partial} \left(\frac{1}{B^2} \right) - \frac{\partial \phi}{\partial} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right) \right) \sum_{z \neq 0}^{n_s-1} n + \\ & + \sum_{=0}^{n_s-1} \left(\frac{4}{3} \eta \left(\frac{\partial}{h \partial} \parallel \right)^2 + \eta_{AN} \left(\frac{\partial}{h_y \partial y} \parallel \right)^2 \right) + \dots + \dots \end{aligned}$$

Contribution from friction between charged species

Old form

$$\dots = \sum_{=0}^{n_s-1} \sum_{z \neq 0}^{n_s-1} \frac{1}{4 \tau_p} \sqrt{\frac{\dots}{\dots + \dots}} z^2 z_{\bullet}^2 n_{\bullet} \left(\parallel - \dots \right)^2$$

New form is used when b2sigp_style is 2

$$\dots = - \sum_{=0}^{n_s-1} \parallel \left(\dots_f + \dots_e \right)$$

Atom heat source due to ionization

$$\begin{aligned} \dots &= \sum_{=0}^{n_s-2} \dots, \\ \dots &= \begin{cases} n_e n \left(\frac{+}{4} +1 \right) \left(\parallel - \parallel_{+1} \right)^2, & z < z_{+1} \text{ \& } z_n = z_{n+1} \\ 0 \end{cases} \end{aligned}$$

Atom heat source due to recombination

$$\begin{aligned} \left(\right) &= \sum_{i=1}^{n_s-1} \left(\right) \\ \left(\right) &= \begin{cases} \frac{n_s n_s}{4} \left(\frac{+}{-1} \right) \left(\parallel - \parallel_{-1} \right)^2, & z_{-1} < z \quad \& \quad z_{n-1} = z_n \\ 0 \end{cases} \end{aligned}$$

$$\tau_p = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{\frac{p}{\Lambda}}^{\frac{3}{2}}}{\Lambda} \left(\frac{4\pi\epsilon_0}{e^2} \right)^2, \text{ where the Coulomb logarithm } \Lambda \text{ can be chosen as}$$

$$\Lambda = \begin{cases} 12 \\ \begin{cases} 23,4 - 1,15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) & 3,45 \cdot \log_{10} \left(\frac{e}{e} \right), & \frac{e}{e} \leq 50 \\ 25,3 - 1,15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) & 2,30 \cdot \log_{10} \left(\frac{e}{e} \right), & \frac{e}{e} > 50 \end{cases} & \text{Braginsky formula} \\ \max \left\{ 5; 17,3 - 0,5 \cdot \log_e \left(\frac{n_e}{10^{20}} \right) \right. & \left. 1,5 \cdot \log_e \left(\frac{e}{e \cdot 10^3} \right) \right\} & \text{Wesson formula for ion-ion collisions} \end{cases},$$

$$\tilde{q} = \tilde{f} - \kappa \frac{1}{\hbar} \frac{\partial}{\partial} - q^{P. \hbar}$$

$$\tilde{f} = \frac{3}{2} \Gamma + \left(\right) n$$

$$q^{P. \hbar} = \frac{5}{2} \sum_{\substack{=0 \\ z \neq 0}}^{n_s-1} D_y \frac{1}{\hbar} \frac{\partial n}{\partial y}^2$$

$$q = \left[\frac{3}{2} \Gamma + \frac{5}{2} \sum_{\substack{=0 \\ z \neq 0}}^{n_s-1} n \sim^{(d)} + \left(\right) n \right] - \kappa \frac{1}{\hbar} \frac{\partial}{\partial}$$

$$\Gamma = \sum_{=0}^{n_s-1} \Gamma^{(\hbar e)}$$

$$q^{(E)} = \frac{5}{2} \sum_{\substack{=0 \\ z \neq 0}}^{n_s-1} \Gamma^{(E)} - \kappa \frac{1}{\hbar} \frac{\partial}{\partial}$$

$$\kappa = \kappa^{(AN)} + \tilde{\kappa}^{(CL)},$$

$$\left(AN \right) = \sum_{=0}^{n_s-1} \sim^{(AN)},$$

$$\sim_{(AN)} = \left\{ \begin{array}{ll} {}^{(AN)}, & z \neq 0 \\ \frac{{}^{(AN)}}{\left(1+\frac{\left(\frac{\sqrt{\mathfrak{s}}}{\mathfrak{h}}\frac{{}^{(AN)}\left|\frac{\partial}{\partial}\right|}{\frac{\sqrt{\mathfrak{s}}}{\mathfrak{h}}\alpha_{\mathfrak{h}}n}\right)^{\gamma_2}\right)^{\frac{1}{\gamma_2}}}, & z=0 \end{array} \right.$$

$$\mathfrak{h}=\sqrt{2}$$

$${}^{(AN)}=\chi^{(AN)}n\; ,$$

$$\frac{10^{20}}{.2} \qquad .3 \frac{6,2 \; 10^{-2}}{\left| \right|} \qquad (\; , \;),$$

$$c^{(l_u\ c\ n)} = \begin{cases} \frac{1}{1 + \frac{7,5 \cdot 10^{16} \left(\frac{-}{e} \right)^2}{c(y) \sum_{n=0}^{n_s-1} z^2 n}}, & \text{for } nne\ f_u\ \& f\ ces \\ 1, & \text{for } othe\ f_u\ \& f\ ces \end{cases}$$

For plateau and banana correction ‘b2trcl_lluciani’ ‘3’

$$c^{(l_u\ c\ n)} = 1.6 \cdot \epsilon^{3/2} / ,$$

$$^2 = \frac{0.66 + 1.88 \cdot \sqrt{\epsilon} - 1.54 \cdot \epsilon}{1 + 1.03 \cdot \nu_*^{1/2} + 0.31 \cdot \nu_*} + \frac{1.17 \cdot \epsilon^3 \cdot \nu_*}{1 + 0.74 \cdot \epsilon^{3/2} \cdot \nu_*}$$

$$\kappa^{(CL)} = \frac{5}{2} b^2 \sum_{\substack{n=0 \\ z \neq 0}}^{n_s-1} n\ f\ \tau\ ,$$

$$f\ = 2.253$$

For the integral modeling (‘b2trcl_integral_modeling’ ‘1’) in the region of the overlap with ASTRA in the inner part of the core:

$$\kappa^{(CL)} = \frac{25}{4} \left(n_l\ \frac{B_z}{B_y} \right)^2 \frac{y(q_{te}\ dp\ ne)}{e} \left[\oint \frac{\sqrt{s}}{B^2} \left(1 - \frac{B^2}{\langle B^2 \rangle} \right)^2 d \right] \times A\ \underline{\hspace{1cm}}$$

$$\text{here } \langle B^2 \rangle = \frac{\oint B^2 \sqrt{s} d}{\oint \sqrt{s} d}$$

from ASTRA we get and $\oint q^{NEO} d$ as a functions of $\underline{\hspace{1cm}}$ and calculate

$$A\ \underline{\hspace{1cm}} = \frac{-1}{\oint q^{NEO} d} \frac{d()}{d(\underline{\hspace{1cm}})} \text{ as a function of the flux surface in B2}$$

$$^{(\tau)} = 0$$

$$\tilde{q}_y = \tilde{f}_y - \kappa_y \frac{1}{b_y} \frac{\partial}{\partial y} - q_y^{P.\ b}$$

$$\tilde{f}_y = \frac{3}{2} \Gamma_y + ^{(\tau)} n + \frac{3}{2} \frac{\left(j^{(AN)} + \tilde{j}_y^{(v\ sll)} + j_y^{(n)} \right)}{e}$$

$$q_y^{P.\ b} = - \frac{5}{2} \sum_{\substack{n=0 \\ z \neq 0}}^{n_s-1} D_y \frac{1}{b_y} \frac{\partial n}{\partial}^2$$

$$q_y=\left[\frac{3}{2}\Gamma_y+\frac{5}{2}\sum_{\substack{=0\\z\neq 0}}^{n_s-1}n\,\,\tilde{\sim}_y^{(d\,\,)}+\,\,\tilde{\sim}_y^{(\,\,\iota\,\,)}n+\frac{3}{2}\beta_{\text{Scott}}\frac{j_y^{(n)}}{e}+\frac{3}{2}\frac{\left(j^{(AN)}_y+\tilde{j}_y^{(v\,\text{sl})}+j^{(n)}_y\right)}{e}\right]-\kappa_y\boldsymbol{\rlap{\hbox{\scriptsize $\boldsymbol{\mathfrak{h}}$}}}_y\frac{1}{\partial y}$$

$$\beta_{\text{Scott}}=\begin{cases}0, & \text{for } u\neq s\\1, & \text{for } B_{u\neq s}\text{ ce Scott}\end{cases}$$

$$\Gamma_y=\sum_{=0}^{n_s-1}\Gamma_y^{(\boldsymbol{\mathfrak{h}}\,e)}$$

$$q_y^{(E\,\,)}=\frac{5}{2}\sum_{\substack{=0\\z\neq 0}}^{n_s-1}\Gamma_y^{(E\,\,)}\,\,\,-\kappa_y\boldsymbol{\rlap{\hbox{\scriptsize $\boldsymbol{\mathfrak{h}}$}}}_y\frac{1}{\partial y}$$

$$\tilde{\sim}_y^{(\,\,\iota\,\,)}=0$$

$$\kappa_y=\kappa_y^{(AN)}+\kappa_y^{(CL)},$$

$$\tilde{\sim}_y^{(AN)}=\sum_{=0}^{n_s-1}\tilde{\sim}_y^{(AN)},$$

$$\tilde{\sim}_y^{(AN)}=\begin{cases}\tilde{\sim}_y^{(AN)}, & z\neq 0\\ \frac{\tilde{\sim}_y^{(AN)}}{\left(1+\left(\frac{\frac{\sqrt{\mathcal{S}}}{\boldsymbol{\rlap{\hbox{\scriptsize $\boldsymbol{\mathfrak{h}}$}}}_y}\left|\tilde{\sim}_y^{(AN)}\frac{\partial}{\partial y}\right|\right)^{\gamma_2}\right)^{\frac{1}{\gamma_2}}}, & z=0\end{cases}$$

$$\mathbf{\kappa}_y^{(CL)}=0$$

$$\frac{1}{\sqrt{\mathcal{S}}}\frac{\partial}{\partial}\left(\frac{\sqrt{\mathcal{S}}}{\boldsymbol{\rlap{\hbox{\scriptsize $\boldsymbol{\mathfrak{h}}$}}}}j\right)+\frac{1}{\sqrt{\mathcal{S}}}\frac{\partial}{\partial y}\left(\frac{\sqrt{\mathcal{S}}}{\boldsymbol{\rlap{\hbox{\scriptsize $\boldsymbol{\mathfrak{h}}$}}}_y}j_y\right)=0$$

$$j\,\,=\dot{j}^{(AN)}+\tilde{j}^{(d\,\,)}+j^{(n)}+\tilde{j}^{(v\,\text{sl})}+\tilde{j}^{(v\,s\perp)}+\tilde{j}^{(v\,sq)}+\tilde{j}^{(s)}+j^{(\text{ll})},$$

$$j^{(AN)}=-\sigma_{AN}\boldsymbol{\rlap{\hbox{\scriptsize $\boldsymbol{\mathfrak{h}}$}}}\frac{\partial\phi}{\partial}$$

$$\tilde{j}^{(d\,\,)}=B_z\frac{\sum_{\substack{=0\\z\neq 0}}^{n_s-1}n\,\,(z_e+\,\,)}{\boldsymbol{\rlap{\hbox{\scriptsize $\boldsymbol{\mathfrak{h}}$}}}_y}\frac{\partial}{\partial y}\left(\frac{1}{B^2}\right),$$

$$j^{(n)} = \frac{B_z}{2\hbar_y} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right) \sum_{\substack{=0 \\ z \neq 0}}^{n_s-1} n^2, \quad z = 1,$$

$$\tilde{j}^{(v\parallel)} = \begin{cases} -\frac{\hbar_z B}{3B^2 \hbar_y} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right) \sum_{\substack{=0 \\ z \neq 0}}^{n_s-1} \eta \frac{\partial(\sqrt{B})}{\hbar \partial}, & z = 1, \\ 0, & z \neq 1, \end{cases},$$

$$\tilde{j}^{(v\perp)} = 0,$$

$$\tilde{j}^{(vsq)} = -0.24\alpha^{NEO} \tau \hbar_z \frac{B}{B^2 \hbar_y} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right) \frac{\partial \left(q^{(0)} B^{\frac{1}{2}} \right)}{\hbar \partial}, \quad z = 1 \quad \text{nd} \quad z_n = 1, \quad q^{(0)} \text{ see above}$$

$$\tilde{j}^{(s)} = -\sigma_0 \hbar_z^2 \frac{\partial \phi}{\hbar \partial} - \sigma_0 \hbar_z^2 \frac{1}{Z_{en}} \frac{\partial h}{\hbar \partial} + \sigma_0 B_{z_{0y}}, \quad z = 1, \quad ?$$

$$\sigma_0 = \frac{n \langle \sigma_e \rangle n_0}{2B^2}, \quad z = 1, ?$$

$$\langle \sigma_e \rangle = 3.2 \cdot 10^{-15} \sqrt{0.026e}$$

$$_{0y} = \frac{\Gamma_{0y}}{n_0}$$

$$j^{(\parallel)} - \text{see above}$$

$$j_y = j_y^{(AN)} + \tilde{j}_y^{(d)} + j_y^{(n)} + \tilde{j}_y^{(v\parallel)} + j_y^{(v\perp)} + \tilde{j}_y^{(vsq)} + \tilde{j}_y^{(s)} + j_y^{(\cdot)},$$

$$j_y^{(AN)} = -\sigma_{AN} \hbar_y \frac{\partial \phi}{\partial y},$$

$$\tilde{j}_y^{(d)} = -\frac{B_z \sum_{\substack{=0 \\ z \neq 0}}^{n_s-1} n (z_e +)}{\hbar} \frac{\partial}{\partial} \left(\frac{1}{B^2} \right),$$

$$j_y^{(n)} = -\frac{B_z}{2\hbar} \frac{\partial}{\partial} \left(\frac{1}{B^2} \right) \sum_{\substack{=0 \\ z \neq 0}}^{n_s-1} n^2,$$

$$\tilde{j}_y^{(v\parallel)} = \begin{cases} \boldsymbol{b}_z \frac{B}{3B^2} \frac{\partial}{\partial} \left(\frac{1}{B^2} \right) \sum_{\substack{n_s=1 \\ z \neq 0}} \eta \frac{\partial \left(B^{\frac{1}{2}} \right)}{\partial}, & z = 1, \text{ , not agreed with parallel momentum balance?} \\ 0, & z \neq 1, \end{cases}$$

$$j_y^{(v\perp)} = -\frac{1}{B\sqrt{\mathcal{S}}}\frac{\partial}{\partial y}\left(\frac{\sqrt{\mathcal{S}}}{\mathcal{H}_y}\frac{\eta_{AN}}{4}\frac{\partial}{\mathcal{H}_y\partial y}\right), \quad =_{\text{main}}$$

$$_{\perp} = -\frac{1}{B\mathcal{H}_y}\frac{\partial \phi}{\partial y} - \frac{1}{z\,eB\mathcal{H}_yn}\frac{\partial n}{\partial y}$$

$$\tilde{j}_y^{(vsq)} = 0.24\alpha^{NEO}\tau\boldsymbol{b}_z\frac{B}{B^2}\frac{\partial}{\mathcal{H}_y\partial y}\left(\frac{1}{B^2}\right)\frac{\partial\left(q^{(0)}B^{\frac{1}{2}}\right)}{\mathcal{H}_y\partial}, \quad z=1 \quad nd \quad z_n=1, \; q^{(0)} \text{ see above}$$

$$\tilde{j}_y^{(s)} = -\sigma_0\frac{\partial\phi}{\mathcal{H}_y\partial y} - \sigma_0\frac{1}{Zen}\frac{\partial n}{\mathcal{H}_y\partial y} - \sigma_0B_{z=0}, \quad =1, \; ?$$

$$_0 = \frac{\Gamma_0}{n_0}$$

$$j_y^{(\quad)} = \begin{cases} -\sigma^{(\quad)}\left(\frac{1}{\mathcal{H}_y}\frac{\partial\phi}{\partial y} - \frac{e}{en_e\mathcal{H}_y}\frac{\partial n_e}{\partial y} - \frac{0.5}{e\mathcal{H}_y}\frac{\partial}{\partial y}\right), & y\in[-\Delta^{(\quad)},0] \\ 0, & y\notin[-\Delta^{(\quad)},0] \end{cases}$$

$$\sigma^{(\quad)} = \begin{cases} c_{\sigma^{(\quad)}}en_e, & y\in[-\Delta^{(\quad)},0] \\ 0, & y\notin[-\Delta^{(\quad)},0] \end{cases}, \qquad c_{\sigma^{(\quad)}} > c_{\sigma_{ANy}}$$

$$\sigma_y = \sigma_{AN} + \sigma_{CLy}$$

$$\sigma_{AN} \text{ depends on 'b2tqna_model_sig'}$$

$$\sigma_{AN} = \left(c_{\sigma,0} + c_{\sigma,1} \frac{10^{20}}{n_e} + c_{\sigma,3} \frac{6,2 \cdot 10^{-2}}{e|B|} e + c_{\sigma,4} e^{-c_{\sigma,5} \left(\frac{y}{L(\quad)} \right)^2} \right) en_e,$$

or

$$\sigma_{AN} = \left(c_{\sigma,0} + c_{\sigma,1} \frac{10^{20}}{n_e} + c_{\sigma,3} \frac{6,2 \cdot 10^{-2}}{e|B|} e + c_{\sigma,4} e^{-c_{\sigma,5} \left(\frac{y}{L(\quad)} \right)^2} \right) e\tilde{n}_e, \; \tilde{n}_e - dens\; ty \; t\; nne \; f_u \; \& \; f_{ce}$$

$$L(\quad) = \int\limits_0^{y_{\Phi_i}} \mathcal{H}_y(\quad,y) dy$$

$$c_{\sigma,0}=10^{-6}-10^{-3},\, c_{\sigma,1}=0,\, c_{\sigma,3}=0,$$
$$\sigma_{CL_y}=0$$

Typical boundary conditions

i. Particle balance equation

$$n = n_{e,0} + n_{e,1}n$$

South. Private region

$$z = 0$$

$$\left. \frac{\sqrt{\mathcal{E}}}{\mathcal{L}_y} \Gamma_y \right|_0 = \left[p_{n,1} \frac{\sqrt{\mathcal{E}}}{\mathcal{L}_y} \sqrt{z_e +} n \right]_0 + \sum_{ec,0}^{n_s-1} c_{ns} \max \left\{ 0, -\frac{\sqrt{\mathcal{E}}}{\mathcal{L}_y} \Gamma_y \right\} \Bigg|_0 \quad \text{leakage, BCCON=10}$$

$$p_{n,1} = -10^{-2}, \quad c_{ec,0,0} = 0, \quad c_{ec,1,0} = 1$$

$\tilde{\Gamma}_y$ was replaced by Γ_y

c_{ns} -neutral sources rescale coefficient. 1 – when working B2 and 10^{-10} when eirene coupling

$$z \neq 0$$

BCCON=9 is below

$$\left. \frac{\sqrt{\mathcal{E}}}{\mathcal{L}_y} \Gamma_y \right|_0 = \begin{cases} \left[-\frac{1}{p_{n,1}} \frac{\sqrt{\mathcal{E}}}{\mathcal{L}_y} (D_{AN}^n + D_{AN}^P (z_e +)) n \right]_0, & df_{fnb} = 0 \\ \left[-\frac{1}{p_{n,1}} \frac{\sqrt{\mathcal{E}}}{\mathcal{L}_y} (D_{AN}^n + D_{AN}^P (z_e +)) n + \frac{\sqrt{\mathcal{E}}}{\mathcal{L}_y} D_y \frac{1}{\mathcal{L}_y} \frac{\partial n}{\partial} + \frac{\sqrt{\mathcal{E}}}{\mathcal{L}_y} n^{(E)} \right]_0, & df_{fnb} = 1 \end{cases}$$

$$\left. \frac{no}{n_{e,0}} \frac{df}{df} \right|_0 = 0$$

$$\left. \frac{no}{n_{e,1}} \frac{df}{df} \right|_0 = -\frac{1}{p_{n,1}} \frac{\sqrt{\mathcal{E}}}{\mathcal{L}_y} (D_{AN}^n + D_{AN}^P (z_e +))$$

BCCON=10 (leakage) is below

$$\left. \frac{\sqrt{\mathcal{E}}}{\mathcal{L}_y} \tilde{\Gamma}_y \right|_0 = \begin{cases} \left[p_{n,1} \frac{\sqrt{\mathcal{E}}}{\mathcal{L}_y} \sqrt{z_e +} n + \min \left\{ 0, \frac{\sqrt{\mathcal{E}}}{\mathcal{L}_y} n^{(E)} \right\} \right]_0, & df_{fnb} = 0 \\ \left[p_{n,1} \frac{\sqrt{\mathcal{E}}}{\mathcal{L}_y} \sqrt{z_e +} n + \frac{\sqrt{\mathcal{E}}}{\mathcal{L}_y} D_y \frac{1}{\mathcal{L}_y} \frac{\partial n}{\partial} + \min \left\{ 0, \frac{\sqrt{\mathcal{E}}}{\mathcal{L}_y} n^{(E)} \right\} \right]_0, & df_{fnb} = 1 \end{cases}$$

$$p_{n,1} = -10^{-3}$$

$$\frac{dn}{dz}\bigg|_{z=0} = 0$$

$$\frac{dn}{dz}\bigg|_{z=1} = p_{n,1} \frac{\sqrt{\mathcal{E}}}{h_y} \sqrt{z_e + n}$$

BCCON=15 is below

$$-\frac{\sqrt{\mathcal{E}}}{h_y^2} D_y \frac{\partial n}{\partial y} \bigg|_0 = \left[c_{n,2} \frac{\sqrt{\mathcal{E}}}{h_y} \sqrt{z_e + n} \right]_0$$

$$c_{n,2} = -10^{-3}$$

South 1. Core plasma

$$z = 0$$

$$\frac{\sqrt{\mathcal{E}}}{h_y} \Gamma_y \bigg|_1 = 0 \quad \text{total particle flux (8), dpc}$$

$$\frac{\sqrt{\mathcal{E}}}{h_y} \tilde{\Gamma}_y \bigg|_1 = 0$$

$$z \neq 0$$

constant density ()

$$\frac{\sqrt{\mathcal{E}}}{h_y} \tilde{\Gamma}_y \bigg|_1 = \frac{\sqrt{\mathcal{E}}}{h_y} (v_{hs} n|_1 - v_{hs} n) \bigg|_1, \quad v_{hs} \approx 10^{12}, \quad n|_1 \text{ is a given density}$$

the total particle flux with constant flux density (8)

$$\frac{\sqrt{\mathcal{E}}}{h_y} \tilde{\Gamma}_y \bigg|_1 = p_{n,0} \frac{\frac{\sqrt{\mathcal{E}}}{h_y}}{\sum \left(\frac{\sqrt{\mathcal{E}}}{h_y} \right)_{-1}} + \frac{\sqrt{\mathcal{E}}}{h_y} D_y \frac{1}{h_y} \frac{\partial n}{\partial} + \frac{\sqrt{\mathcal{E}}}{h_y} {}^{(E)} n \bigg|_1$$

North

$$z = 0$$

$$-\frac{\sqrt{s}}{\hbar_y} \Gamma_y \Big|_N = \left[p_{n, \cdot, 1} \frac{\sqrt{s}}{\hbar_y} \sqrt{z_e +} n \Big|_N + \sum_{=0}^{n_s-1} c_{ec, \cdot, 0} c_{ns} \max \left\{ 0, \frac{\sqrt{s}}{\hbar_y} \Gamma_y \right\} \right]_{=0}^{n_s-1} \quad e \cdot s e (0), \text{dpc}$$

$$p_{n, \cdot, 1} = 0, \quad c_{ec, 0, 0} = 0, \quad c_{ec, 1, 0} = 1$$

$$-\frac{\sqrt{s}}{\hbar_y} \Gamma_y \Big|_N = \sum_{=0}^{n_s-1} c_{ec, \cdot, 0} c_{ns} \max \left\{ 0, \frac{\sqrt{s}}{\hbar_y} \Gamma_y \right\} \quad \Gamma_y \text{ was replaced by } \Gamma_y$$

$$c_{ec, 0, 0} = 0, \quad c_{ec, 1, 0} = 1$$

$$z \neq 0$$

BCCON=9 is below

$$-\frac{\sqrt{s}}{\hbar_y} \Gamma_y \Big|_N = \begin{cases} \left[\frac{1}{p_{n, \cdot, 1}} \frac{\sqrt{s}}{\hbar_y} (D_{AN}^n + D_{AN}^p (z_e +)) n \right]_0, & df_{-fn} = 0 \\ \left[\frac{1}{p_{n, \cdot, 1}} \frac{\sqrt{s}}{\hbar_y} (D_{AN}^n + D_{AN}^p (z_e +)) n - \frac{\sqrt{s}}{\hbar_y} D_y \frac{1}{\hbar_y} \frac{\partial n}{\partial} - \frac{\sqrt{s}}{\hbar_y} {}^{(E)}_y n \right]_N, & df_{-fn} = 1 \end{cases}$$

$$no_{n, \cdot, 0}^{df} = 0$$

$$no_{n, \cdot, 1}^{df} = \frac{1}{p_{n, \cdot, 1}} \frac{\sqrt{s}}{\hbar_y} (D_{AN}^n + D_{AN}^p (z_e +))$$

BCCON=10 (leakage) is below

$$-\frac{\sqrt{s}}{\hbar_y} \Gamma_y \Big|_N = \begin{cases} \left[p_{n, \cdot, 1} \frac{\sqrt{s}}{\hbar_y} \sqrt{z_e +} n - \max \left\{ 0, \frac{\sqrt{s}}{\hbar_y} n {}^{(E)}_y \right\} \right]_N, & df_{-fn} = 0 \\ \left[p_{n, \cdot, 1} \frac{\sqrt{s}}{\hbar_y} \sqrt{z_e +} n - \frac{\sqrt{s}}{\hbar_y} D_y \frac{1}{\hbar_y} \frac{\partial n}{\partial} - \max \left\{ 0, \frac{\sqrt{s}}{\hbar_y} n {}^{(E)}_y \right\} \right]_N, & df_{-fn} = 1 \end{cases}$$

$$no_{n, \cdot, 0}^{df} = 0$$

$$no_{n, \cdot, 1}^{df} = p_{n, \cdot, 1} \frac{\sqrt{s}}{\hbar_y} \sqrt{z_e +} n$$

$$\left. \frac{\sqrt{\mathcal{S}}}{h_y^2} D_y^n \frac{\partial n}{\partial y} \right|_N = \left[c_{n,2} \frac{\sqrt{\mathcal{S}}}{h_y} \sqrt{z_e + n} \right]_N \quad \text{BCCON=15}$$

$$c_{n,2} = -10^{-3}$$

North with puffing

$$z = 0$$

$$\left. -\frac{\sqrt{\mathcal{S}}}{h_y} \tilde{\Gamma}_y \right|_N = p_{n,0} \frac{\sqrt{\mathcal{S}}}{h_y} \Big|_N + \sum_{=0}^{n_s-1} \left[c_{ec,0} c_{ns} \max \left\{ 0, \frac{\sqrt{\mathcal{S}}}{h_y} \Gamma_y \right\} \right]_N \quad \text{the particle flux per unit area (5)}$$

or

$\tilde{\Gamma}_y$ was replaced by Γ_y

$$\left. -\frac{\sqrt{\mathcal{S}}}{h_y} \tilde{\Gamma}_y \right|_N = \frac{p_{n,0} \frac{\sqrt{\mathcal{S}}}{h_y}}{\sum_{-0}^{N-1} \left(\frac{\sqrt{\mathcal{S}}}{h_y} \right)_{,N_y}} + \sum_{=0}^{n_s-1} \left[c_{ec,0} c_{ns} \max \left\{ 0, \frac{\sqrt{\mathcal{S}}}{h_y} \Gamma_y \right\} \right]_N \quad \text{the total particle flux with constant}$$

flux density (8)

$\tilde{\Gamma}_y$ was replaced by Γ_y

$$c_{ec,0,0} = 0, c_{ec,1,0} = 1, p_{n,0,0} \sim 10^{21},$$

West

$$z = 0$$

$$\left. \frac{\sqrt{\mathcal{S}}}{h_y} \Gamma \right| = \left[p_{n,1} \frac{\sqrt{\mathcal{S}}}{h_y} + \sum_{=0}^{n_s-1} c_{ec,0} c_{ns} \max \left\{ 0, -\frac{\sqrt{\mathcal{S}}}{h_y} \Gamma \right\} \right] \quad \text{particle flux per area (5), dpc}$$

$$p_{n,1} = 0, c_{ec,0,0} = 0, c_{ec,1,0} = 1$$

$$\left. \frac{\sqrt{\mathcal{S}}}{h_y} \tilde{\Gamma} \right| = \left[p_{n,1} \frac{\sqrt{\mathcal{S}}}{h_y} + \sum_{=0}^{n_s-1} c_{ec,0} c_{ns} \max \left\{ 0, -\frac{\sqrt{\mathcal{S}}}{h_y} \Gamma \right\} \right] \quad \text{It has been replaced } \tilde{\Gamma} \text{ by } \Gamma. \text{ (5)}$$

$$p_{n,1} = 0, c_{ec,0,0} = 0, c_{ec,1,0} = 1$$

$$z \neq 0$$

$$\left. \frac{\sqrt{s}}{h} \Gamma \right| = 10^{20} \frac{\sqrt{s}}{h} \frac{\partial n}{\partial} \quad \text{sheath density bc (3) dpc}$$

$$\left. \frac{\sqrt{s}}{h} \tilde{\Gamma} \right| = \left[-p_{n,1} \frac{\sqrt{s}}{h} \left| h \right. \sqrt{\frac{\sum_{b=0}^{n_s-1} p_b}{\sum_{b=0}^{n_s-1} n_b}} n - \frac{\sqrt{s}}{h} D_y \frac{1}{h_y} \frac{\partial n}{\partial y} \right] \quad \text{sound speed flux bc (14)}$$

$$p_{n,1} = +1,$$

East

$$z = 0$$

$$\left. -\frac{\sqrt{s}}{h} \Gamma \right|_E = \left[p_{n,1} \frac{\sqrt{s}}{h} + \sum_{=0}^{n_s-1} c_{ec,0} c_{ns} \max \left\{ 0, \frac{\sqrt{s}}{h} \Gamma \right\} \right] \quad \text{particle flux per area (5), dpc}$$

$$p_{n,1} = 0, \quad c_{ec,0,0} = 0, \quad c_{ec,1,0} = 1$$

$$\left. -\frac{\sqrt{s}}{h} \tilde{\Gamma} \right|_E = \left[\sum_{=0}^{n_s-1} c_{ec,0} c_{ns} \max \left\{ 0, \frac{\sqrt{s}}{h} \Gamma \right\} \right] \quad \text{It has been replaced } \tilde{\Gamma} \text{ by } \Gamma. (5)$$

$$c_{ec,0,0} = 0, \quad c_{ec,1,0} = 1,$$

$$z \neq 0$$

$$\left. -\frac{\sqrt{s}}{h} \Gamma \right|_E = -10^{20} \frac{\sqrt{s}}{h} \frac{\partial n}{\partial} \quad \text{sheath density bc (3) dpc}$$

$$\left. -\frac{\sqrt{s}}{h} \Gamma \right|_E = \left[-p_{n,1} \frac{\sqrt{s}}{h} \left| h \right. \sqrt{\frac{\sum_{b=0}^{n_s-1} p_b}{\sum_{b=0}^{n_s-1} n_b}} n + \frac{\sqrt{s}}{h} D_y \frac{1}{h_y} \frac{\partial n}{\partial y} \right] \quad \text{sound speed flux bc (14)}$$

$$p_{n,1} = +1$$

ii. **Parallel momentum balance for ions**

$$= \quad_{,0} + \quad_{\parallel} + \quad_{,2} n + \quad_{,3} n u_{\parallel}$$

South. Private region

$$z = 0,$$

$$\left. \frac{\sqrt{\mathcal{S}}}{\mathcal{H}_y} \Gamma_y \right|_0 = \left[p_{,1} p_{,2} \frac{\sqrt{\mathcal{S}}}{\mathcal{H}_y} - p_{,2} \frac{\sqrt{\mathcal{S}}}{\mathcal{H}_y} \right]_{\parallel} \quad \text{weak velocity (4) dpc}$$

$$p_{,1} = 0, p_{,2} = 10^5$$

We had this b.c.

$$\left. \frac{\sqrt{\mathcal{S}}}{\mathcal{H}_y} \Gamma_y \right|_0 = \left[c_{,3} \frac{\sqrt{\mathcal{S}}}{\mathcal{H}_y} n_{\parallel} + c_{,6} \max \left(0, \frac{\sqrt{\mathcal{S}}}{\mathcal{H}_y} \frac{\Gamma_y^{(co)}}{n} \right) n_{\parallel} \right]_0, \quad \text{delete term with max !}$$

$$\Gamma_y^{(co)} = \Gamma_y + \Gamma_y^{(d)}$$

$$c_{,3} = -10^{10}, \text{ we have } c_{,6} = -1, \text{ but not } c_{,6} = 0$$

Inserted into b2stbc_phys for bcmom (2) as $c_{,3} \rightarrow p_{,2} = -10^{10}, c_{,6} \rightarrow p_{,1} = -1$

$$z \neq 0,$$

$$\left. \frac{\sqrt{\mathcal{S}}}{\mathcal{H}_y} \Gamma_y \right|_0 = \left[p_{,1} p_{,2} \frac{\sqrt{\mathcal{S}}}{\mathcal{H}_y} - p_{,2} \frac{\sqrt{\mathcal{S}}}{\mathcal{H}_y} \right]_{\parallel} \quad \text{weak velocity (4) dpc}$$

$$p_{,1} = 0, p_{,2} = 10^5$$

$$\left. \frac{\sqrt{\mathcal{S}}}{\mathcal{H}_y} \Gamma_y \right|_0 = \left[c_{,3} \frac{\sqrt{\mathcal{S}}}{\mathcal{H}_y} n_{\parallel} + c_{,6} \max \left(0, \frac{\sqrt{\mathcal{S}}}{\mathcal{H}_y} \frac{\Gamma_y^{(co)}}{n} \right) n_{\parallel} \right]_0, \quad \text{delete term with max !}$$

$$\Gamma_y^{(co)} = \Gamma_y + \Gamma_y^{(d)}$$

$$c_{,3} = 0, c_{,6} = -1$$

South 1. Core

$$z = 0$$

$$\left. \frac{\sqrt{s}}{y} \Gamma_y \right|_1 = \left[p_{,1} p_{,2} \frac{\sqrt{s}}{y} - p_{,2} \frac{\sqrt{s}}{y} n_{\parallel} \right]_1 \quad \text{weak velocity (4) dpc}$$

$$p_{,1} = 0, p_{,2} = 10^5$$

$$\text{BCMOM}=12 \quad p_{,2} = -10^{10}, p_{,1} = 0$$

$$\left. \frac{\sqrt{s}}{y} \Gamma_y \right|_1 = P_{,2} \frac{\sqrt{s}}{y} n_{\parallel} \Big|_1$$

$$z \neq 0$$

$$\left. \frac{\sqrt{s}}{y} \Gamma_y \right|_1 = \left[p_{,1} p_{,2} \frac{\sqrt{s}}{y} - p_{,2} \frac{\sqrt{s}}{y} n_{\parallel} \right]_1 \quad \text{weak velocity (4) dpc}$$

$$p_{,1} = 0, p_{,2} = 10^5$$

$$\text{BCMOM}=14 \quad p_{,1} = 0, p_{,3} = -10^{10}$$

$$\begin{aligned} \left. \frac{\sqrt{s}}{y} \Gamma_y \right|_1 &= p_{,3} \frac{\sqrt{s}}{y} n_{\parallel} \Big|_1 + \left[\frac{\sqrt{s}}{y} \left\langle n \right\rangle (-2) \frac{\langle B_z \rangle}{ez} \frac{\partial}{\partial} \left(\frac{1}{B^2} \right) \langle n_{\parallel} \rangle c_{dfu} + c_{cofu} \frac{\sqrt{s}}{y} \right]_1 + \\ &+ p_{,1} \max \left(0, \frac{\sqrt{s}}{y} \frac{\Gamma_y^{(co)}}{n} \right) n_{\parallel} \Big|_1 \end{aligned}$$

The value of the parallel velocity as a function of poloidal coordinate

$$\left. \frac{\sqrt{s}}{y} \Gamma_y \right|_1 = v_{bs} n_{e0} \frac{\sqrt{s}}{y} \left[\frac{\langle B \rangle}{B} p_{,1} - n_{\parallel} \right]_1, \quad \text{BCMOM}= 5$$

where $\langle B \rangle = \frac{\oint B \sqrt{s} d}{\oint \sqrt{s} d}$, the integration is done over inner magnetic surface

$$v_{bs} \approx 10^{12}, n_{e0} = 10^{20}, p_{,1} \text{ is a given value of the average parallel velocity}$$

North

$$z = 0,$$

$$\frac{h_z}{h_y} \frac{\sqrt{s}}{h_y} \Gamma_y \Big|_N = \left[p_{,1} p_{,2} \frac{h_z}{h_y} \frac{\sqrt{s}}{h_y} - p_{,2} \frac{h_z}{h_y} \frac{\sqrt{s}}{h_y} \parallel \right]_N \quad \text{weak velocity (4) dpc}$$

$$p_{,1} = 0, p_{,2} = 10^5$$

$$\frac{h_z}{h_y} \frac{\sqrt{s}}{h_y} \Gamma_y \Big|_N = \left[c_{,3} \frac{h_z}{h_y} \frac{\sqrt{s}}{h_y} n_{\parallel} + c_{,6} \max \left(0, \frac{h_z}{h_y} \frac{\sqrt{s}}{h_y} \frac{\Gamma_y^{(co)}}{n} \right) n_{\parallel} \right]_N \quad \text{delete term with max !}$$

$$c_{,3} = -10^{10}, \text{ we have } c_{,6} = -1, \text{ but not } c_{,6} = 0$$

Inserted into b2stbc_phys for BCMOM=12 as $c_{,3} \rightarrow p_{,2} = -10^{10}, c_{,6} \rightarrow p_{,1} = -1$

$$z \neq 0,$$

$$\frac{h_z}{h_y} \frac{\sqrt{s}}{h_y} \Gamma_y \Big|_N = \left[p_{,1} p_{,2} \frac{h_z}{h_y} \frac{\sqrt{s}}{h_y} - p_{,2} \frac{h_z}{h_y} \frac{\sqrt{s}}{h_y} \parallel \right]_N \quad \text{weak velocity (4) dpc}$$

$$p_{,1} = 0, p_{,2} = 10^5$$

$$\frac{h_z}{h_y} \frac{\sqrt{s}}{h_y} \Gamma_y \Big|_N = \left[c_{,3} \frac{h_z}{h_y} \frac{\sqrt{s}}{h_y} n_{\parallel} + c_{,6} \max \left(0, \frac{h_z}{h_y} \frac{\sqrt{s}}{h_y} \frac{\Gamma_y^{(co)}}{n} \right) n_{\parallel} \right]_N$$

$$c_{,3} = 0, c_{,6} = -1$$

West

$$z = 0, p_{,1} = 0$$

$$z \neq 0, p_{,1} = 1$$

$$\frac{h_z}{h_y} \frac{\sqrt{s}}{h_y} \Gamma_y \Big| = \left[p_{,1} \frac{u_{\parallel} h_z}{\epsilon^2} n_{est} - \frac{u_{\parallel} h_z}{\epsilon^2} \frac{h_z}{h_y} \frac{\sqrt{s}}{h_y} \parallel \right] \quad \text{BCMOM=13}$$

$$est = \frac{h_z \sqrt{s} |b|}{h b} \quad (E \times B) \quad \frac{h_z \sqrt{s} b}{h} \sqrt{\frac{\sum_{=0}^{n_s-1} p}{\sum_{=0}^{n_s-1} n}}$$

$z = 0$ *specified parallel velocity (1) dpc*

$$\frac{\sqrt{s}}{h} \Gamma \mid = \left[u_{bs} n_{e00} \quad , \quad \frac{\sqrt{s}}{h} p_{,1} u_{bs} n_{e00} \quad , \quad \frac{\sqrt{s}}{h} \parallel \right] \quad , \quad p_{,1} = 0, u_{bs} = 10^{18}, \quad n_{e00} = 10^{20} \text{ dpc}$$

$z \neq 0$, *sheath velocity (3) dpc*

$p_{,2} < 0.5$, sound_model=0,

$$\frac{\sqrt{s}}{h} \Gamma \mid = \left[-\frac{u_{bs}}{\varepsilon^2} n \quad \frac{\sqrt{s}}{h} u_{p_{q1}} - \frac{u_{bs}}{\varepsilon^2} n \quad \frac{\sqrt{s}}{h} \parallel \right] \quad , \quad \text{dpc}$$

$p_{,1} = 1, \quad p_{,2} = 0, u_{bs} = 10^{18}, \quad n_{e00} = 10^{20}, \quad \varepsilon = 10^{-10}$

$$u_{p_{q1}} = \max \left\{ 0, \frac{u_{u_{q1}}}{b} + \frac{u^{(d)}}{b} \right\}, \quad u_{u_{q1}} = p_{,1} c_s |b|, \quad c_s = \sqrt{\frac{\sum_{=0}^{n_s-1} p}{\sum_{=0}^{n_s-1} n}},$$

East

$z = 0, \quad p_{,1} = 0$

$z \neq 0, \quad p_{,1} = 1$

$$\frac{h_z \sqrt{s}}{h} \Gamma \mid_E = \left[p_{,1} \frac{u_{bs}}{\varepsilon^2} \quad est \quad n \quad - \frac{u_{bs} h_z \sqrt{s}}{\varepsilon^2} \frac{|b|}{h} \mid \quad n \parallel \right]_E \quad \text{BCMOM=13}$$

$$Est = \frac{h_z \sqrt{s} |b|}{h b} \quad (E \times B) \quad + \frac{h_z \sqrt{s} b}{h} \sqrt{\frac{\sum_{=0}^{n_s-1} p}{\sum_{=0}^{n_s-1} n}}$$

$z = 0$, *specified parallel velocity (1) dpc*

$$-\frac{\sqrt{\mathcal{E}}}{\mathcal{L}} \Gamma \Big|_E = \left[u_{\mathcal{B}\mathcal{S}} n_{e00}, \frac{\sqrt{\mathcal{E}}}{\mathcal{L}} p_{\mathcal{S},1} u_{\mathcal{B}\mathcal{S}} n_{e00}, \frac{\sqrt{\mathcal{E}}}{\mathcal{L}} \parallel \right] \Big|_E, \quad p_{\mathcal{S},1} = 0, u_{\mathcal{B}\mathcal{S}} = 10^{18}, \quad n_{e00} = 10^{20} \text{ dpc}$$

$z \neq 0$, *sheath velocity (3) dpc*

$p_{\mathcal{S},2} < 0.5$, sound_model=0,

$$-\frac{\sqrt{\mathcal{E}}}{\mathcal{L}} \Gamma \Big|_E = \left[-\frac{u_{\mathcal{B}\mathcal{S}}}{\mathcal{E}^2} n, \frac{\sqrt{\mathcal{E}}}{\mathcal{L}} u_{p_{\mathcal{B}}}, -\frac{u_{\mathcal{B}\mathcal{S}}}{\mathcal{E}^2} n, \frac{\sqrt{\mathcal{E}}}{\mathcal{L}} \parallel \right] \Big|_E, \text{ dpc}$$

$p_{\mathcal{S},1} = 1, p_{\mathcal{S},2} = 0, u_{\mathcal{B}\mathcal{S}} = 10^{18}, n_{e00} = 10^{20}, \mathcal{E} = 10^{-10}$

$$u_{p_{\mathcal{B}}} = \max \left\{ 0, \frac{u_{u_{\mathcal{B}}}}{\mathcal{B}} - \frac{u^{(d)}}{\mathcal{B}} \right\}, \quad u_{u_{\mathcal{B}}} = p_{\mathcal{S},1} c_s |\mathcal{B}|, \quad c_s = \sqrt{\frac{\sum_{i=0}^{n_s-1} p}{\sum_{i=0}^{n_s-1} n}},$$

iii. Current continuity equation

$$\mathcal{B} = \mathcal{B}_{,0} + \mathcal{B}_{,1} \phi + \mathcal{B}_{,2} n_e + \mathcal{B}_{,3} n_e \phi$$

South. Private region

$$\frac{\sqrt{\mathcal{E}}}{\mathcal{L}} j_y \Big|_0 = 0, \text{ dpc}$$

$$\frac{\sqrt{\mathcal{E}}}{\mathcal{L}} j_y \Big|_0 = \begin{cases} 0, & f_{cd} f_t = 0 \\ \frac{\sqrt{\mathcal{E}}}{\mathcal{L}} \left(\tilde{j}_y^{(d)} + j_y^{(n)} + \tilde{j}_y^{(v\parallel)} + j_y^{(v\perp)} + \tilde{j}_y^{(vsg)} + \tilde{j}_y^{(s)} \right) \Big|_0, & f_{cd} f_t \neq 0 \end{cases}$$

South 1. Core

$$\frac{\sqrt{\mathcal{E}}}{\mathcal{L}} j_y \Big|_1 = \begin{cases} 0, & f_{cd} f_t = 0 \\ \frac{\sqrt{\mathcal{E}}}{\mathcal{L}} \left(\langle \tilde{j}_y^{(d)} \rangle \right) \Big|_0, & f_{cd} f_t \neq 0 \end{cases} \quad \langle \tilde{j}_y^{(d)} \rangle?, \text{ dpc}$$

$$\left.\frac{\sqrt{\mathcal{S}}}{\hbar}j_y\right|_1=\begin{cases}0,& f\text{ }cd\text{ }ft=0\\ \left.\frac{\sqrt{\mathcal{S}}}{\hbar}\left(\left\langle\tilde{j}_y^{(d\text{ })}\right\rangle+\left\langle j_y^{(n)}\right\rangle\right)\right|_0,& f\text{ }cd\text{ }ft\neq0\end{cases}$$

$$\left\langle\tilde{j}_y^{(d\text{ })}\right\rangle=-\frac{B_z\left\langle\sum_{\substack{=0\\z\neq0}}^{n_s-1}n\left(z_e+\right)\right\rangle}{\hbar}\frac{\partial}{\partial}\left(\frac{1}{B^2}\right),$$

$$\left\langle j_y^{(n)}\right\rangle=-\frac{B_z}{2\hbar}\frac{\partial}{\partial}\left(\frac{1}{B^2}\right)\sum_{\substack{=1\\z\neq0}}^{n_s-1}n\left\langle\begin{array}{c}2\\||\end{array}\right\rangle,$$

North

$$\left.\frac{\sqrt{\mathcal{S}}}{\hbar}j_y\right|_N=\begin{cases}0,& f\text{ }cd\text{ }ft=0\\ \left.\frac{\sqrt{\mathcal{S}}}{\hbar}\left(\left\langle\tilde{j}_y^{(d\text{ })}\right\rangle\right)\right|_N,& f\text{ }cd\text{ }ft\neq0\end{cases}\quad\left\langle\tilde{j}_y^{(d\text{ })}\right\rangle?,\textcolor{red}{\text{dpc}}$$

$$-\left.\frac{\sqrt{\mathcal{S}}}{\hbar}j_y\right|_N=\begin{cases}0,& f\text{ }cd\text{ }ft=0\\ -\left.\frac{\sqrt{\mathcal{S}}}{\hbar}\left(\tilde{j}_y^{(d\text{ })}+j_y^{(n)}+\tilde{j}_y^{(v\text{ }s||)}+j_y^{(v\text{ }s\perp)}+\tilde{j}_y^{(v\text{ }sq)}+\tilde{j}_y^{(s)}\right)\right|_N,& f\text{ }cd\text{ }ft\neq0\end{cases}$$

West

BCPOT=11 is connected with BCCON=14 when istyle_fchi = 1

$$\left.\frac{\sqrt{\mathcal{S}}}{\hbar}j\right|=\frac{\sqrt{\mathcal{S}}}{\hbar}e\left[-\sum_{=0}^{n_s-1}z\left(p_{n_s-1}|b\left[\sqrt{\frac{\sum_{b=0}^{n_s-1}p_b}{\sum_{b=0}^{n_s-1}bn_b}}n\right)+(1-\gamma_e)|b\left|n_e\frac{1}{\sqrt{2\pi}}\sqrt{\frac{-e}{e}}\exp\left(-\frac{e\phi}{e}\right)\right|\right],\text{ it is new form}$$

BCPOT=11 when istyle_fchi = 0

$$\left.\frac{\sqrt{\mathcal{S}}}{\hbar}j\right|=\frac{\sqrt{\mathcal{S}}}{\hbar}e\left[\sum_{=0}^{n_s-1}z\left[\Gamma+(1-\gamma_e)|b\left|n_e\frac{1}{\sqrt{2\pi}}\sqrt{\frac{-e}{e}}\exp\left(-\frac{e\phi}{e}\right)\right|\right]\right],\text{ it is old form}$$

sheath potential bc (3)

$$\left.\frac{\sqrt{\mathcal{S}}}{\hbar}j\right|=\frac{\sqrt{\mathcal{S}}}{\hbar}e\left[\sum_{=0}^{n_s-1}z\left[\Gamma+(1-\gamma_e)|b\left|n_e\frac{1}{\sqrt{2\pi}}\sqrt{\frac{-e}{e}}\exp\left(-\frac{e(\varphi-p_{\varphi,2})}{e}\right)\right|\right]\right],\textcolor{red}{\text{dpc}}$$

East

BCPOT=11 is connected with BCCON=14 when istyle_fchi = 1

$$-\frac{\sqrt{\mathcal{E}}}{\hbar} j \Big|_E = \frac{\sqrt{\mathcal{E}}}{\hbar} e \left[- \sum_{n=0}^{n_s-1} z \left(p_{n_s,1} \left| \frac{\sum_{b=0}^{n_s-1} p_b}{\sum_{b=0}^{n_s-1} b n_b} n \right. + (1-\gamma_e) \left| n_e \frac{1}{\sqrt{2\pi}} \sqrt{\frac{-e}{e}} \exp\left(-\frac{e\phi}{e}\right) \right| \right] \right]_E, \text{ it is new form}$$

BCPOT=11 when istyle_fchi = 0

$$-\frac{\sqrt{\mathcal{E}}}{\hbar} j \Big|_E = \frac{\sqrt{\mathcal{E}}}{\hbar} e \left[- \sum_{n=0}^{n_s-1} z \Gamma + (1-\gamma_e) \left| n_e \frac{1}{\sqrt{2\pi}} \sqrt{\frac{-e}{e}} \exp\left(-\frac{e\phi}{e}\right) \right| \right]_E, \text{ it is old form}$$

$$c_{b,7}=1-\gamma_e, \quad c_{b,6}=1$$

sheath potential bc (3)

$$-\frac{\sqrt{\mathcal{E}}}{\hbar} j \Big|_E = \frac{\sqrt{\mathcal{E}}}{\hbar} e \left[- \sum_{n=0}^{n_s-1} z \Gamma + (1-\gamma_e) \left| n_e \frac{1}{\sqrt{2\pi}} \sqrt{\frac{-e}{e}} \exp\left(-\frac{e(\phi-p_{\phi,2})}{e}\right) \right| \right]_E, \text{ dpc}$$

iv. Energy balance equation for electrons

$$\mathfrak{h}_e = \mathfrak{h}_{e,0} + \mathfrak{h}_{e,1} e + \mathfrak{h}_{e,2} n_e + \mathfrak{h}_{e,3} n_{e\ e}$$

South. Private region

$$\frac{\sqrt{\mathcal{E}}}{\hbar} \tilde{q}_{ey} \Big|_0 = \left[\left(a_{e,2} \frac{\sqrt{\mathcal{E}}}{\hbar} \sqrt{\frac{-e}{e}} \right) e n_e + a_{e,7} \max\left(0, -\frac{\sqrt{\mathcal{E}}}{\hbar} \tilde{f}_{ey} \right) e - q_e^{P. \mathfrak{h}} \right]_0, \quad \text{bcene=11}$$

$$a_{e,2} = -10^{-4}, \quad a_{e,7} = -1, \quad a_{e,2} \rightarrow p_{e,2} = -10^{-4}, \quad a_{e,7} \rightarrow p_{e,1} = -1$$

$$-\frac{\sqrt{\mathcal{E}}}{\hbar} e y \frac{\partial}{\hbar} \frac{\partial}{\partial y} \Big|_0 = \left[\left(p_{e,1} \frac{\sqrt{\mathcal{E}}}{\hbar} \sqrt{\frac{-e}{e}} \right) e n_e \right]_0, \quad \text{bcene=22}$$

BCENE=9 (electron temperature decay length)

$$\frac{\sqrt{t}}{2} \left[\begin{array}{c} -\frac{\sqrt{t}}{2} + - \\ -\frac{\sqrt{t}}{2} + - \Sigma \end{array} \right] =$$

$$-\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}\tilde{q}_{ey}|_N=\begin{cases}\left[\frac{5}{2}\sum_{=0}^{n_s-1}z_{no_{n,0}}^{df_e}-\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}\frac{\kappa_e^{(AN)}}{p_{e,1}}\right]_e+\frac{5}{2}\sum_{=0}^{n_s-1}z_{no_{n,1}}^{df_n_e}\right]_N, & df_{\gamma}e=0 \\ \left[\frac{5}{2}\sum_{=0}^{n_s-1}z_{no_{n,0}}^{df_e}-\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}\frac{\kappa_e^{(AN)}}{p_{e,1}}\right]_e+\frac{5}{2}\sum_{=0}^{n_s-1}z_{no_{n,1}}^{df_n_e}+q_{ey}^{P,\textcolor{blue}{b}}-\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}\frac{3}{2}\frac{(E)}{ey}n_{e_e}\right]_N, & df_{\gamma}e=1\end{cases}$$

West

bcene_15_style.eq.0 is connected with BCCON=14

$$\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}\tilde{q}_e|_e=-\left(\frac{1+\gamma}{1-\gamma}e+e\varphi\right)\max\left(0,\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}\sum_{=0}^{n_s-1}z_{p_{n,1}}|b|\sqrt{\frac{\sum_{b=0}^{n_s-1}p_b}{\sum_{b=0}^{n_s-1}bn_b}}n+\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}\frac{\tilde{j}_{\parallel}}{e}\right)-\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}q_e^{P,\textcolor{blue}{b}}\right] \textcolor{blue}{bcene=15}$$

$$\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}\tilde{q}_e|_e=-\left(\frac{1+\gamma}{1-\gamma}+\frac{e\varphi}{e}\right)\max\left(0,\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}\sum_{=0}^{n_s-1}z_{p_{n,1}}|b|\sqrt{\frac{\sum_{b=0}^{n_s-1}p_b}{\sum_{b=0}^{n_s-1}bn_b}}n+\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}\frac{\tilde{j}_{\parallel}}{e}\right)_e-\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}q_e^{P,\textcolor{blue}{b}}\right] \textcolor{blue}{bcene=15}$$

bcene_15_style.eq.1

$$\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}\tilde{q}_e|_e=\left[-\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}|b|\left((1+\gamma)_e+(1-\gamma)e\varphi\right)n_e\frac{1}{\sqrt{2\pi}}\sqrt{\frac{e}{e}}\exp\left(-\frac{e\phi}{e}\right)-\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}q_e^{P,\textcolor{blue}{b}}\right] \textcolor{blue}{bcene=15}$$

sheath electron (3)

$$p_{n,2}<0.5, \text{ sound_mode1}=0,$$

$$f_{p_{e,1}}>0 \nrightarrow en$$

$$\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}q_e|_e=\left[p_{e,1}+e\frac{\varphi-p_{\varphi,2}}{e}\right]\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}v_{e_{q_i}}-\frac{\sqrt{\delta_{\gamma}}}{h_{\gamma}}q_e^{P,\textcolor{blue}{b}}\Bigg|, \quad p_{e,1}=0.9, \quad p_{\varphi,2}=0,$$

$$v_{e_{q_i}}=\min\left\{0,-u_{u_{q_i}}n_e-\frac{j}{e}\right\}, \quad u_{u_{q_i}}=p_{n,1}c_s|b|, \quad c_s=\sqrt{\frac{\sum_{=0}^{n_s-1}p}{\sum_{=0}^{n_s-1}n}},$$

end f

East

bcene_15_style.eq.0 is connected with BCCON=14

$$-\frac{\sqrt{\mathcal{S}}}{h} \tilde{q}_e \Big|_E = \left[- \left(\frac{1+\gamma}{1-\gamma} e + e\varphi \right) \max \left(0, \frac{\sqrt{\mathcal{S}}}{h} \sum_{b=0}^{n_s-1} z_{p_{n_s,1}} |b| \sqrt{\frac{\sum_{b=0}^{n_s-1} p_b}{\sum_{b=0}^{n_s-1} b n_b}} n - \frac{\sqrt{\mathcal{S}}}{h} \frac{\tilde{j}_{\parallel}}{e} \right) + \frac{\sqrt{\mathcal{S}}}{h} q_e^{P, b} \right]_E \quad \text{bcene=15}$$

$$-\frac{\sqrt{\mathcal{S}}}{h} \tilde{q}_e \Big|_E = \left[- \left(\frac{1+\gamma}{1-\gamma} + \frac{e\varphi}{e} \right) \max \left(0, \frac{\sqrt{\mathcal{S}}}{h} \sum_{b=0}^{n_s-1} z_{p_{n_s,1}} |b| \sqrt{\frac{\sum_{b=0}^{n_s-1} p_b}{\sum_{b=0}^{n_s-1} b n_b}} n - \frac{\sqrt{\mathcal{S}}}{h} \frac{\tilde{j}_{\parallel}}{e} \right) e + \frac{\sqrt{\mathcal{S}}}{h} q_e^{P, b} \right]_E \quad \text{bcene=15}$$

bcene_15_style.eq.1

$$-\frac{\sqrt{\mathcal{S}}}{h} \tilde{q}_e \Big|_E = \left[- \frac{\sqrt{\mathcal{S}}}{h} |b| \left((1+\gamma) e + (1-\gamma) e\varphi \right) n_e \frac{1}{\sqrt{2\pi}} \sqrt{\frac{e}{e}} \exp \left(- \frac{e\varphi}{e} \right) + \frac{\sqrt{\mathcal{S}}}{h} q_e^{P, b} \right]_E \quad \text{bcene=15}$$

sheath electron (3)

$p_{s,2} < 0.5$, sound_model=0,

$f_{p_{s,1}} > 0$ h en

$$-\frac{\sqrt{\mathcal{S}}}{h} \tilde{q}_e \Big|_E = - \left[p_{s,1} + e \frac{\varphi - p_{\varphi,2}}{e} \right] \frac{\sqrt{\mathcal{S}}}{h} v_{e_{q_i}} e + \frac{\sqrt{\mathcal{S}}}{h} q_e^{P, b} \Big|_E, \quad p_{s,1} = 0.9, \quad p_{\varphi,2} = 0,$$

$$v_{e_{q_i}} = \max \left\{ 0, u_{u_{q_i}} n_e - \frac{j}{e} \right\}, \quad u_{u_{q_i}} = p_{s,1} c_s |b|, \quad c_s = \sqrt{\frac{\sum_{b=0}^{n_s-1} p}{\sum_{b=0}^{n_s-1} n}},$$

end f

v. Energy balance equation for ions

$$h = h_{s,0} + h_{s,1} + h_{s,2} n + h_{s,3} n$$

South. Private region

$$p_{\bullet,1} = 10^{-2}, e_{ec,0} = 0, e_{ec,1} = 0.3, e_{ec,0,0} = 0, e_{ec,1,0} = 1$$

BCENE=9 (electron temperature decay length)

$$\frac{\sqrt{\delta^*}}{h_y} \tilde{q}_y \Big|_0 = \begin{cases} \left[\frac{5}{2} \sum_{n=0}^{n_s-1} \frac{no_{n,0}^{df}}{n} - \frac{\sqrt{\delta^*}}{h_y} \frac{\kappa^{(AN)}}{p_{\bullet,1}} + \frac{5}{2} \sum_{n=1}^{n_s-1} \frac{no_{n,1}^{df}}{n} + \right. \\ \left. + \sum_{n=0}^{n_s-1} \max \left\{ e_{ec}, \left(\frac{3}{2} + B_{\alpha} \right) \left(\frac{z_{ef} + z_{ef}^+}{z_{ef} + z_{ef}^+} \right), \left(\frac{3}{2} + B_{\alpha} \right) \right\} e_{ec,0} c_{ns} \max \left\{ 0, -\frac{\sqrt{\delta^*}}{h_y} \Gamma_y \right\} \right]_0, & df_{-h} = 0 \\ \left[\frac{5}{2} \sum_{n=0}^{n_s-1} \frac{no_{n,0}^{df}}{n} - \frac{\sqrt{\delta^*}}{h_y} \frac{\kappa^{(AN)}}{p_{\bullet,1}} + \frac{5}{2} \sum_{n=1}^{n_s-1} \frac{no_{n,1}^{df}}{n} - q_y^{p.b} + \frac{\sqrt{\delta^*}}{h_y} \frac{3}{2} \sum_{n=0}^{n_s-1} \frac{(E)}{y} n + \right. \\ \left. + \sum_{n=0}^{n_s-1} \max \left\{ e_{ec}, \left(\frac{3}{2} + B_{\alpha} \right) \left(\frac{z_{ef} + z_{ef}^+}{z_{ef} + z_{ef}^+} \right), \left(\frac{3}{2} + B_{\alpha} \right) \right\} e_{ec,0} c_{ns} \max \left\{ 0, -\frac{\sqrt{\delta^*}}{h_y} \Gamma_y \right\} \right]_0, & df_{-h} = 1 \end{cases}$$

$$\frac{\sqrt{\delta^*}}{h_y} \tilde{q}_y \Big|_0 = p_{\bullet,2} \sum_{\substack{n=0 \\ z \neq 0}}^{n_s-1} \left(\frac{\sqrt{\delta^*}}{h_y} n \sqrt{\frac{z_{ef} + z_{ef}^+}{z_{ef} + z_{ef}^+}} \right) \Big|_0 + p_{\bullet,1} \max \left(0, -\frac{\sqrt{\delta^*}}{h_y} \tilde{f}_y \right) \Big|_0, \quad \text{bceni}=21$$

$$p_{\bullet,1} = -1, p_{\bullet,2} = -10^{-2}$$

$$-\frac{\sqrt{\delta^*}}{h_y} \frac{\partial}{\partial y} \Big|_0 = p_{\bullet,1} \sum_{\substack{n=0 \\ z \neq 0}}^{n_s-1} \left(\frac{\sqrt{\delta^*}}{h_y} n \sqrt{\frac{z_{ef} + z_{ef}^+}{z_{ef} + z_{ef}^+}} \right) \Big|_0, \quad \text{bceni}=22$$

$$\frac{\sqrt{\delta^*}}{h_y} \tilde{q}_y \Big|_0 = \sum_{\substack{n=0 \\ z \neq 0}}^{n_s-1} \left(a_{\bullet,2} \frac{\sqrt{\delta^*}}{h_y} \frac{n}{n} \sqrt{\frac{z_{ef} + z_{ef}^+}{z_{ef} + z_{ef}^+}} \right) n \Big|_0 + a_{\bullet,0,7} \max \left(0, -\frac{\sqrt{\delta^*}}{h_y} \tilde{f}_y \right) \Big|_0, \quad \text{b2stbc_spb}$$

$$a_{\bullet,2} = -10^{-2}$$

South 1. Core

tot on ene sy fu (8)

$$\left. \frac{\sqrt{\mathcal{E}}}{h_y} \tilde{q}_y \right|_1 = p_{\cdot,1} \frac{\frac{\sqrt{\mathcal{E}}}{h_y}}{\sum_{coe} \left(\frac{\sqrt{\mathcal{E}}}{h_y} \right)} \Bigg|_1, \quad p_{\cdot,1} = 8 \cdot 10^5, \quad \text{dpc}$$

$$\left. \frac{\sqrt{\mathcal{E}}}{h_y} \tilde{q}_y \right|_1 = \sum_{=0}^{n_s-1} \left[\left(c_{\cdot,0} \frac{\sqrt{\mathcal{E}}}{h_y} \frac{n}{n} + c_{\cdot,1} \frac{\sqrt{\mathcal{E}}}{h_y} \frac{n}{n} \right) \right] \Bigg|_1$$

$$c_{\cdot,0} = 4 \cdot 10^{13}, c_{\cdot,1} = -10^{30}, \quad = 0, 1,$$

North

BCENE=9 (electron temperature decay length)

$$p_{\cdot,1} = 10^{-2}, e_{ec,0} = 0, e_{ec,1} = 0.3, \quad e_{ec,0,0} = 0, \quad e_{ec,1,0} = 1$$

$$-\left. \frac{\sqrt{\mathcal{E}}}{h_y} \tilde{q}_y \right|_N = \begin{cases} \left[\frac{5}{2} \sum_{=0}^{n_s-1} \frac{no_{\cdot,0}^{df}}{n} - \frac{\sqrt{\mathcal{E}}}{h_y} \frac{\kappa^{(AN)}}{p_{\cdot,1}} + \frac{5}{2} \sum_{=0}^{n_s-1} \frac{no_{\cdot,1}^{df}}{n} n + \right. \\ \left. + \sum_{=0}^{n_s-1} \max \left\{ e_{ec}, \left(\frac{3}{2} + B_{\cdot} \right) \left(\frac{z_{\cdot} + \cdot}{\cdot} \right), \left(\frac{3}{2} + B_{\cdot} \right) \right\} e_{ec,0} c_{ns} \max \left\{ 0, -\frac{\sqrt{\mathcal{E}}}{h_y} \Gamma_y \right\} \right]_N, & df_{\cdot} = 0 \\ \left[\frac{5}{2} \sum_{=0}^{n_s-1} \frac{no_{\cdot,0}^{df}}{n} - \frac{\sqrt{\mathcal{E}}}{h_y} \frac{\kappa^{(AN)}}{p_{\cdot,1}} + \frac{5}{2} \sum_{=0}^{n_s-1} \frac{no_{\cdot,1}^{df}}{n} n + q_y^{P,\cdot} - \frac{\sqrt{\mathcal{E}}}{h_y} \frac{3}{2} \sum_{=0}^{n_s-1} \frac{(E)}{y} n + \right. \\ \left. + \sum_{=0}^{n_s-1} \max \left\{ e_{ec}, \left(\frac{3}{2} + B_{\cdot} \right) \left(\frac{z_{\cdot} + \cdot}{\cdot} \right), \left(\frac{3}{2} + B_{\cdot} \right) \right\} e_{ec,0} c_{ns} \max \left\{ 0, -\frac{\sqrt{\mathcal{E}}}{h_y} \Gamma_y \right\} \right]_N, & df_{\cdot} = 1 \end{cases}$$

$$-\left. \frac{\sqrt{\mathcal{E}}}{h_y} \tilde{q}_y \right|_N = p_{\cdot,2} \sum_{\substack{=0 \\ z \neq 0}}^{n_s-1} \left(\frac{\sqrt{\mathcal{E}}}{h_y} n \sqrt{\frac{z_{\cdot} + \cdot}{\cdot}} \right) \Bigg|_N + p_{\cdot,1} \max \left(0, -\frac{\sqrt{\mathcal{E}}}{h_y} \tilde{f}_y \right) \Bigg|_N,$$

bceni=21

$$p_{\cdot,1} = -1, p_{\cdot,2} = -10^{-2}$$

$$\left. \frac{\sqrt{\mathcal{E}}}{h_y} \frac{\partial}{h_y \partial y} \right|_N = p_{\cdot,1} \sum_{\substack{=0 \\ z \neq 0}}^{n_s-1} \left(\frac{\sqrt{\mathcal{E}}}{h_y} n \sqrt{\frac{z_{\cdot} + \cdot}{\cdot}} \right) \Bigg|_N,$$

bceni=22

$$-\frac{\sqrt{s}}{h_y}\tilde{q}_y|_N=\sum_{\substack{=0\\z\neq 0}}^{n_s-1}\left(a_{\substack{,2}}\frac{\sqrt{s}}{h_y}\frac{n}{n}\sqrt{\frac{z_e+}{}}\right)n\left|+a_{\substack{,0,7}}\max\left(0,-\frac{\sqrt{s}}{h_y}\tilde{f}_y\right)\right|_N,$$

b2stbc_spb

$a_{\substack{,2}}=-10^{-2}$

West

$$\frac{\sqrt{s}}{h}\tilde{q}|=-p_{\substack{,1}}\frac{\sqrt{s}}{h}|b|\sqrt{\frac{\sum_{b=0}^{n_s-1}p_b}{\sum_{b=0}^{n_s-1}bn_b}\sum_{\substack{=0\\z\neq 0}}^{n_s-1}p_{n_{\substack{,1}}n}-\frac{\sqrt{s}}{h}q^{p_{\substack{,1}}b}}}$$

BCENI=15 (associated with BCCON=14)

$p_{\substack{,1}}=+1.5$

sheath ion (3)

BCENI=3

$p_{\substack{,n,2}}=0<0.5,$ sound_model=0,

$$\frac{\sqrt{s}}{h}\tilde{q}|=\left[\sum_{z\neq 0}\left(\frac{\sqrt{s}}{h}\frac{p_{\substack{,2}}}{2}u_{u_{q_i}}\frac{n}{n}\left(u_{p_{q_i}}^2+u_{\substack{(d)}}^{(d)}\right)^2\right)-\frac{\sqrt{s}}{h}p_{\substack{u_{q_i}}}u_{u_{q_i}}n\right)+$$

$$+\sum_{=0}^{n_s-1}\max\left\{c_{ec,\substack{,1}}\left(\frac{3}{2}+B_{\substack{,1}}\right)\left(\frac{z_{\substack{f,f,f,f}}}{\substack{,f}}+\substack{,f}\right),\left(\frac{3}{2}+B_{\substack{,1}}\right)\substack{,f,p}\right\}c_{ec,\substack{,0}}c_{ns}\max\left\{0,-\frac{\sqrt{s}}{h}\Gamma\right\}\right]$$

$u_{\substack{(d)}}^{(d)}=u_{\substack{(E)}}^{(E)}+\tilde{u}_{\substack{(d)}}^{(d)}$

$u_{p_{q_i}}=\max\left\{0,\frac{u_{u_{q_i}}}{|b|}+\frac{|B_z|}{|B|}u_{\substack{1,}}^{(d)}\right\}$

$$u_{u_{q_i}}=p_{\substack{,n,1}}c_s|b|,\quad c_s=\sqrt{\frac{\sum_{=0}^{n_s-1}p}{\sum_{=0}^{n_s-1}n}},\quad p_{\substack{,n,1}}=1.0$$

if neutrals_namelist.eq.1 .or. use_eirene.ne.0 then $e_{ec,0}=0$, $e_{ec,1}=0.3$, $e_{ec,0}=0$, $e_{ec,1}=1$ else $c_{ec,0,0}=0$,
 $c_{ec,1,0}=1$, $c_{ec,0,1}=0$, $c_{ec,1,1}=0.3$

$p_{\substack{,1}}=2.5$ but it should be $p_{\substack{,1}}=1.5$

$p_{\substack{,2}}=0$ without drifts

$p_{t,2} = 10^{-3}$ with drifts but it should be $p_{t,2} = 0$

East

$$-\frac{\sqrt{s}}{h} \tilde{q} \Big|_E = -p_{t,1} \frac{\sqrt{s}}{h} \Big|_E \sqrt{\frac{\sum_{k=0}^{n_s-1} p_k}{\sum_{k=0}^{n_s-1} n_k}} \sum_{\substack{n=0 \\ z \neq 0}}^{n_s-1} p_{n,1} n - \frac{\sqrt{s}}{h} q^{p, k} \Big|_E$$

BCENI=15(associated with BCCON=14)

$p_{t,1} = +1.5$

sheath ion (3) BCENI=3

$p_{t,n,2} = 0 < 0.5$, sound_model=0,

$$-\frac{\sqrt{s}}{h} \tilde{q} \Big|_E = \left[\sum_{z \neq 0} \left(-\frac{\sqrt{s}}{h} \frac{p_{t,2}}{2} u_{u_{q,t}} n \left(u_{p_{q,t}}^2 + u_{(d)}^{(d)} \right)^2 \right) - \frac{\sqrt{s}}{h} p_{t,u_{q,t}} n \right] + \left[\sum_{n=0}^{n_s-1} \max \left\{ c_{ec,1} \left(\frac{3}{2} + B_{\text{osc}} \right) \left(\frac{z_{f,q,t}}{u_{f,p}} + \dots \right), \left(\frac{3}{2} + B_{\text{osc}} \right) \dots \right\} c_{ec,0} c_{ns} \max \left\{ 0, \frac{\sqrt{s}}{h} \Gamma \right\} \right] \Big|_E$$

$u^{(d)} = u^{(E)} + \tilde{u}^{(d)}$

$u_{p_{q,t}} = \max \left\{ 0, \frac{u_{u_{q,t}}}{|h|} - \frac{|B_z|}{B} u_{1,}^{(d)} \right\}$

$u_{u_{q,t}} = p_{t,n,1} c_s |h|, \quad c_s = \sqrt{\frac{\sum_{k=0}^{n_s-1} p_k}{\sum_{k=0}^{n_s-1} n}}, \quad p_{t,n,1} = 1.0$

if neutrals_namelist.eq.1 .or. use_eirene.ne.0 then $e_{ec,0} = 0, e_{ec,1} = 0.3, e_{ec,0} = 0, e_{ec,1} = 1$ else $c_{ec,0,0} = 0, c_{ec,1,0} = 1, c_{ec,0,1} = 0, c_{ec,1,1} = 0.3$

$p_{t,1} = 2.5$ but it should be $p_{t,1} = 1.5$

$p_{t,2} = 0$ without drifts

$p_{t,2} = 10^{-3}$ with drifts but it should be $p_{t,2} = 0$

Appendix

Old form of $(\nabla \pi^\parallel)_\parallel$ was

$$(\nabla \pi^\parallel)_\parallel = \begin{cases} 0, & \neq 1 \\ -\frac{4}{3} \frac{b}{B^2} \frac{\partial}{\partial} \left(\frac{c_{\eta^{(CL)}} c_{\eta^{(CL)}}^{(k_c n)} \eta_{CL}}{B^2} \frac{\partial \left(B^{\frac{1}{2}} \left(\parallel + 0 \cdot \frac{B}{B}^{(d)} + 0 \cdot \frac{B}{B}^{(E)} \right) \right)}{\partial} \right) - \frac{0 \cdot B^{\frac{3}{2}}}{\partial} \frac{\partial}{\partial} \left(\frac{b}{v B^2} \frac{\partial \left(B^{\frac{1}{2}} \left(q_\parallel^{(0)} + \frac{B}{B} q^{(d)} \right) \right)}{\partial} \right) \end{cases}, \quad =1$$

Contribution from anomalous viscosity was contained in l.h.s.

Because of $\eta_{AN} \ll \tilde{\eta}_{CL}$, $\eta \approx \tilde{\eta}_{CL}$ and taking into account $B \eta_y = const$

$$-\frac{4}{3} \frac{b}{B^2} \frac{\partial}{\partial} \left(\frac{c_{\eta^{(CL)}} c_{\eta^{(CL)}}^{(k_c n)} \eta_{CL}}{B^2} \frac{\partial \left(B^{\frac{1}{2}} \parallel \right)}{\partial} \right) = \frac{\partial}{\partial} \left(\frac{b}{z \sqrt{s}} \left(\frac{4}{3} \eta \frac{\partial \ln \frac{b}{z}}{\partial} \parallel - \frac{4}{3} \eta \frac{\partial \parallel}{\partial} \right) \right) - \frac{\partial}{\partial} \left(\frac{b}{z \sqrt{s}} \frac{4}{3} \eta \frac{\partial \ln \left(\frac{b}{z} B^{\frac{1}{2}} \right)}{\partial} \right) B^{\frac{1}{2}} \parallel$$

New form of $(\nabla \pi^\parallel)_\parallel$

$$(\nabla \pi^\parallel)_\parallel = \begin{cases} 0, & \neq 1, \\ \frac{\partial}{\partial} \left(\frac{b}{z \sqrt{s}} \left(\frac{4}{3} \eta \frac{\partial \ln \frac{b}{z}}{\partial} \parallel - \frac{4}{3} \eta \frac{\partial \parallel}{\partial} \right) \right) - \frac{\partial}{\partial} \left(\frac{b}{z \sqrt{s}} \frac{4}{3} \eta \frac{\partial \ln \left(\frac{b}{z} B^{\frac{1}{2}} \right)}{\partial} \right) B^{\frac{1}{2}} \parallel \end{cases}, \quad =1$$

About scaling of residuals

$$\frac{\sum_{=0}^n \sum_{j=0}^{-1n_y-1} \left| n_{\text{ , , } j} \right|}{\sum_{=0}^n \sum_{j=0}^{-1n_y-1} \sqrt{\mathcal{S}_{\text{ , } j}} n_{\text{ , , } j}},$$
$$\frac{\sum_{=0}^n \sum_{j=0}^{-1n_y-1} \left| u_{\text{ , , } j} \right|}{\sum_{=0}^n \sum_{j=0}^{-1n_y-1} \sqrt{\mathcal{S}_{\text{ , } j}} n_{\text{ , , } j} \sqrt{\text{---}}}$$
$$\frac{\sum_{=0}^n \sum_{j=0}^{-1n_y-1} \left| \mathfrak{h}_{e\text{ , , } j} \right|}{\frac{3}{2} \sum_{=0}^{n_s-1n} \sum_{=0}^{-1n_y-1} \sum_{j=0}^{\text{---}} \sqrt{\mathcal{S}_{\text{ , } j}} n_{\text{ , , } j} (z_e + \text{---})}$$
$$\frac{\sum_{=0}^n \sum_{j=0}^{-1n_y-1} \left| \mathfrak{h}_{\text{ , , } j} \right|}{\frac{3}{2} \sum_{=0}^{n_s-1n} \sum_{=0}^{-1n_y-1} \sum_{j=0}^{\text{---}} \sqrt{\mathcal{S}_{\text{ , } j}} n_{\text{ , , } j} (z_e + \text{---})}$$
$$\frac{\sum_{=0}^n \sum_{j=0}^{-1n_y-1} \left| po_{\text{ , , } j} \right|}{\sum_{=0}^{n_s-1n} \sum_{=0}^{-1n_y-1} \sum_{j=0}^{\text{---}} \sqrt{\mathcal{S}_{\text{ , } j}} z_{en_{\text{ , , } j}}}$$